

The Lambert W Function

Real Branches, Calculus, Series, Asymptotics,
Approximations, Algorithms, and Applications

Prepared as a self-contained mathematical monograph

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Abstract

The Lambert W function is the inverse, in a multivalued sense, of the map

$$w \mapsto we^w.$$

This deceptively short definition produces a function with two real branches, infinitely many complex branches, an algebraic square-root singularity at $-1/e$, logarithmic behavior at infinity and at the origin of every nonprincipal branch, rich Taylor and Puiseux series, useful continued fractions, and a large collection of applications.

This article develops the subject from first principles. The real branches W_0 and W_{-1} are treated in parallel whenever possible. Every main elementary result is proved, including the real branch structure, derivative and higher-derivative formulas, curvature, local power series, branch-point expansions, all-order real asymptotic expansions, branch-aware inequalities, globally convergent logarithmic Newton iterations, and the standard reductions of exponential-polynomial equations to W . Numerical approximations are organized by the geometric region in which they are useful: the origin, the branch point, large positive arguments, and small negative arguments on W_{-1} . Applications include labeled trees, delay differential equations, iterated exponentials, enzyme kinetics, diode circuits, and estimates for prime numbers.

The attached source archive was used as a literature base. In particular, the canonical analysis of Corless, Gonnet, Hare, Jeffrey, and Knuth, the real-branch approximations of Barry and collaborators, the bounds of Chatzigeorgiou, the continued fractions of Corcino, Corcino, and Mező, the approximation work of Iacono–Boyd and Howard, the guaranteed iteration of Lóczi, and the application papers in circuits, biochemistry, ecology, and prime-number theory are incorporated and cross-referenced. All figures and numerical tables can be regenerated by the included Python program.

Preface and reading guide

The intended reader has completed a standard two-semester calculus sequence and has seen elementary complex numbers, power series, and the inverse function theorem. A little real analysis is helpful but not required. Whenever a more advanced theorem is used, it is stated explicitly and the role it plays is isolated.

There are four natural ways to read the book.

1. For a first encounter, read [Chapters 1 to 3](#), then the first halves of [Chapters 4 to 6](#).
2. For real numerical work, continue with [Chapters 8 to 10](#).
3. For complex analysis, focus on [Sections 1.4 and 6.4](#) and the contour and Fourier representations in [Chapter 7](#).
4. For applications, the chapters in Part IV can be read largely independently after [Chapter 3](#).

A recurring principle is worth stating at the beginning:

The formula alone rarely determines the correct branch. The range, monotonicity, and physical or mathematical constraints of the original problem must be used.

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Part I

Foundations

Chapter 1

Definition, geometry, and branches

1.1 Why a new inverse function is needed

The equation

$$we^w = z$$

contains the unknown both outside and inside an exponential. Taking logarithms only changes the form:

$$w + \log w = \log z,$$

and does not isolate w . This is the same situation in which the logarithm itself arose: elementary algebra does not invert the exponential, so one names and studies its inverse. Here the inverse of a product of the identity and the exponential is called the Lambert W function, also known historically as the product logarithm.

Definition 1.1 (Lambert W relation). For $z \in \mathbb{C}$, a number $w \in \mathbb{C}$ is a value of the Lambert W relation at z when

$$we^w = z.$$

The single-valued analytic inverse branches are denoted by $W_k(z)$, where $k \in \mathbb{Z}$. The branch W_0 is the principal branch.

The adjective “relation” is important. Just as the complex logarithm has infinitely many values, the inverse of we^w has infinitely many complex values. On the real axis, however, only two branches occur.

1.2 The two real branches

Let

$$f(w) = we^w, \quad w \in \mathbb{R}.$$

Then

$$f'(w) = e^w(1 + w), \quad f''(w) = e^w(2 + w).$$

The sign of f' gives the complete real picture.

Theorem 1.2 (Real branch theorem). *The function $f(w) = we^w$ has a unique global minimum at $w = -1$, with*

$$f(-1) = -\frac{1}{e}.$$

It is strictly decreasing from 0^- to $-1/e$ on $(-\infty, -1]$, and strictly increasing from $-1/e$ to $+\infty$ on $[-1, \infty)$. Consequently:

- (a) W_0 is a real-valued increasing function on $[-1/e, \infty)$, with range $[-1, \infty)$;
- (b) W_{-1} is a real-valued decreasing function on $[-1/e, 0)$, with range $(-\infty, -1]$;
- (c) for $-1/e < x < 0$, the equation $we^w = x$ has exactly two real solutions, $W_0(x)$ and $W_{-1}(x)$;
- (d) at $x = -1/e$, the two real branches meet at -1 , and for $x < -1/e$ there is no real solution.

Proof. Because $e^w > 0$, the sign of $f'(w)$ is the sign of $1 + w$. Thus $f'(w) < 0$ for $w < -1$, $f'(-1) = 0$, and $f'(w) > 0$ for $w > -1$. Hence f decreases up to -1 and increases after -1 . Also

$$\lim_{w \rightarrow -\infty} we^w = 0^-, \quad f(-1) = -1/e, \quad \lim_{w \rightarrow \infty} we^w = \infty.$$

The stated ranges follow from continuity and strict monotonicity. A continuous strictly monotone function has a single-valued inverse on its range, giving the two inverse branches and all four conclusions. $\square$

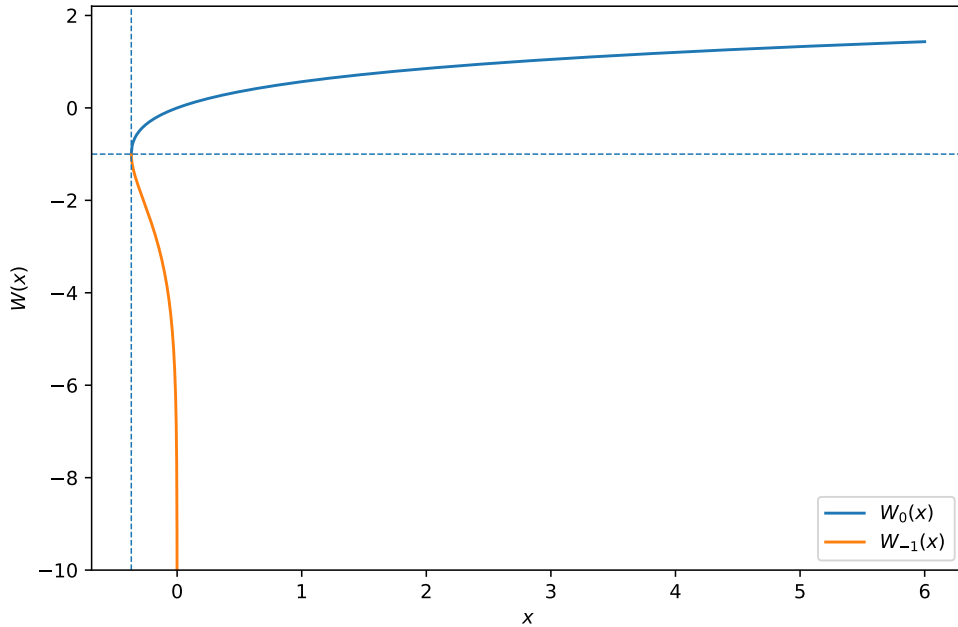


Figure 1.1: The two real branches. The principal branch is increasing; the lower branch is decreasing and tends to $-\infty$ as $x \rightarrow 0^-$.

Remark 1.3 (The nonprincipal real branch is not secondary). Many expositions introduce W_{-1} as an afterthought. This is a mistake in applications. Whenever an equation has two admissible real solutions, the second one is usually carried by W_{-1} . Examples include $x^x = a$ for a range of a , threshold equations, escape-time equations, delay spectra, and asymptotic inversion in prime-number theory.

1.3 Basic values and limits

The defining equation immediately gives a large family of exact values.

Proposition 1.4 (Values generated from the inverse relation). *For every real $a \geq -1$,*

$$W_0(ae^a) = a.$$

For every real $a \leq -1$,

$$W_{-1}(ae^a) = a.$$

In particular,

$$W_0(0) = 0, \quad W_0(-1/e) = W_{-1}(-1/e) = -1, \quad W_0(e) = 1.$$

Moreover,

$$\lim_{x \rightarrow \infty} W_0(x) = \infty, \quad \lim_{x \rightarrow 0^-} W_{-1}(x) = -\infty.$$

Proof. If $x = ae^a$, then a is a solution of $we^w = x$. The range conditions in [Theorem 1.2](#) select the branch. The limits follow by inverting the endpoint limits of f on the corresponding monotonicity intervals. $\square$

The number

$$\Omega = W_0(1) = 0.567143290409783872999968 \dots$$

is called the *omega constant*. It is the unique positive solution of $\Omega e^\Omega = 1$, equivalently $\Omega = e^{-\Omega}$.

1.4 Complex branches in one page

The function $f(w) = we^w$ is entire. Its derivative vanishes only at $w = -1$, so every local inverse is analytic except where continuation meets the critical value $-1/e$, or where the inverse escapes to infinity. The latter phenomenon occurs at $z = 0$ on every nonprincipal branch.

A standard branch convention labels the infinitely many solutions so that, for large z ,

$$W_k(z) \approx \text{Log } z + 2\pi i k - \text{Log}(\text{Log } z + 2\pi i k).$$

This formula explains the integer branch index: changing k changes the logarithm by one turn around the origin. The principal branch W_0 is analytic at 0, whereas W_k for $k \neq 0$ has logarithmic divergence there. The point $-1/e$ is a square-root branch point joining W_0 to W_{-1} . Detailed cut conventions are discussed in the canonical paper of Corless et al. [5]; for real-variable work, [Theorem 1.2](#) removes all ambiguity.

Table 1.1: Representative real values, generated by the included Python program.

argument	branch	value
$-1/e$	W_0	-1.0
$-1/e$	W_{-1}	-1.0
-0.35	W_0	-0.716638816456074
-0.35	W_{-1}	-1.34971725219225
-0.1	W_0	-0.111832559158963
-0.1	W_{-1}	-3.5771520639573
-10^{-6}	W_0	$-1.0000010000015e - 6$
-10^{-6}	W_{-1}	-16.6265089013725
0	W_0	0.0
0.1	W_0	0.0912765271608623
1	W_0	0.567143290409784
e	W_0	1.0
10	W_0	1.7455280027407
10^6	W_0	11.3833580861401

Chapter 2

Calculus of W_0 and W_{-1}

2.1 First derivative and monotonicity

Let W denote any differentiable branch at a point where $W \neq -1$. Differentiate $W(x)e^{W(x)} = x$:

$$W'(x)e^{W(x)}(1 + W(x)) = 1.$$

Theorem 2.1 (Derivative formulas). *At every regular point of a branch,*

$$W'(x) = \frac{e^{-W(x)}}{1 + W(x)}, \quad (2.1)$$

$$= \frac{W(x)}{x(1 + W(x))}, \quad x \neq 0. \quad (2.2)$$

For the principal branch, $W'_0(0) = 1$.

Proof. The first formula follows from implicit differentiation. For $x \neq 0$, the identity $e^{-W} = W/x$, obtained from $We^W = x$, gives the second formula. At the origin, $f'(0) = 1$, so the inverse function theorem gives $W'_0(0) = 1/f'(0) = 1$. Equivalently, take the limit in (2.1). $\square$

The signs now agree with the geometric argument:

$$W'_0(x) > 0 \quad (x > -1/e), \quad W'_{-1}(x) < 0 \quad (-1/e < x < 0).$$

Both derivatives become unbounded in magnitude at the branch point.

2.2 Second derivative and curvature

Differentiating (2.1) once more gives

$$W''(x) = -\frac{e^{-2W(x)}(W(x) + 2)}{(1 + W(x))^3}. \quad (2.3)$$

Theorem 2.2 (Concavity and the inflection point of W_{-1}). *The principal branch W_0 is strictly concave on $(-1/e, \infty)$. The branch W_{-1} is strictly convex on*

$$\left(-\frac{1}{e}, -\frac{2}{e^2}\right)$$

and strictly concave on

$$\left(-\frac{2}{e^2}, 0\right).$$

Its unique inflection point is

$$\left(-\frac{2}{e^2}, -2\right).$$

Proof. On W_0 , one has $W > -1$, so the denominator in (2.3) is positive, while $W + 2 > 1$. Hence $W_0'' < 0$.

On W_{-1} , the denominator is negative. Therefore the sign of W'' is the sign of $W + 2$: it is positive when $-2 < W < -1$, zero at $W = -2$, and negative when $W < -2$. The corresponding argument at $W = -2$ is $-2e^{-2}$. Since W_{-1} is decreasing, the stated interval order follows. $\square$

2.3 Higher derivatives

Repeated differentiation has a useful uniform structure.

Theorem 2.3 (Derivative polynomials). *Define polynomials P_n by*

$$P_1(w) = 1, \quad P_{n+1}(w) = (1 + w)P_n'(w) - (nw + 3n - 1)P_n(w).$$

Then, for $n \geq 1$,

$$\frac{d^n}{dx^n} W(x) = \frac{e^{-nW(x)} P_n(W(x))}{(1 + W(x))^{2n-1}}. \quad (2.4)$$

The first polynomials are

$$P_1 = 1, \quad P_2 = -(w + 2), \quad P_3 = 2w^2 + 8w + 9,$$

$$P_4 = -(6w^3 + 36w^2 + 79w + 64).$$

At the origin,

$$P_n(0) = W_0^{(n)}(0) = (-n)^{n-1}.$$

Proof. The case $n = 1$ is (2.1). Assume (2.4). Since

$$\frac{d}{dx} = \frac{e^{-W}}{1 + W} \frac{d}{dW},$$

we obtain

$$\begin{aligned} \frac{d^{n+1}W}{dx^{n+1}} &= \frac{e^{-W}}{1 + W} \frac{d}{dW} \left(e^{-nW} P_n(W) (1 + W)^{-(2n-1)} \right) \\ &= \frac{e^{-(n+1)W}}{(1 + W)^{2n+1}} \left((1 + W)P_n'(W) - (nW + 3n - 1)P_n(W) \right), \end{aligned}$$

which is the required recurrence. The displayed polynomials follow by direct calculation. The value at zero will also follow from the Taylor series in Theorem 4.3; alternatively, the recurrence at $w = 0$ proves it by induction after the same coefficient identity is established. $\square$

2.4 Differential equations

The derivative identity can be written without exponentials:

$$x(1 + W)W' = W. \quad (2.5)$$

Thus every scaled function $w(x) = W(cx)$ solves the same first-order nonlinear differential equation. Differentiating (2.5) gives a second-order equation,

$$x(1 + W)W'' + x(W')^2 + WW' = 0. \quad (2.6)$$

These identities are often more convenient than the original inverse relation in symbolic calculations.

2.5 Conditioning

If the input x is perturbed by a small relative amount, the first-order relative response of W is measured by

$$\kappa_W(x) = \left| \frac{xW'(x)}{W(x)} \right|.$$

Using (2.2) gives an unusually simple answer.

Proposition 2.4 (Relative condition number). *For either real branch away from the branch point and from the removable point $x = 0$ of W_0 ,*

$$\kappa_W(x) = \frac{1}{|1 + W(x)|}.$$

Near $-1/e$,

$$\kappa_W(x) \sim \frac{1}{\sqrt{2(1 + ex)}}.$$

This holds after choosing either real branch.

Proof. The exact formula follows immediately from (2.2). The branch-point expansion in [Theorem 5.1](#) gives $1 + W_0 \sim p$ and $1 + W_{-1} \sim -p$, where $p = \sqrt{2(1 + ex)}$. Taking absolute values gives the asymptotic form. $\square$

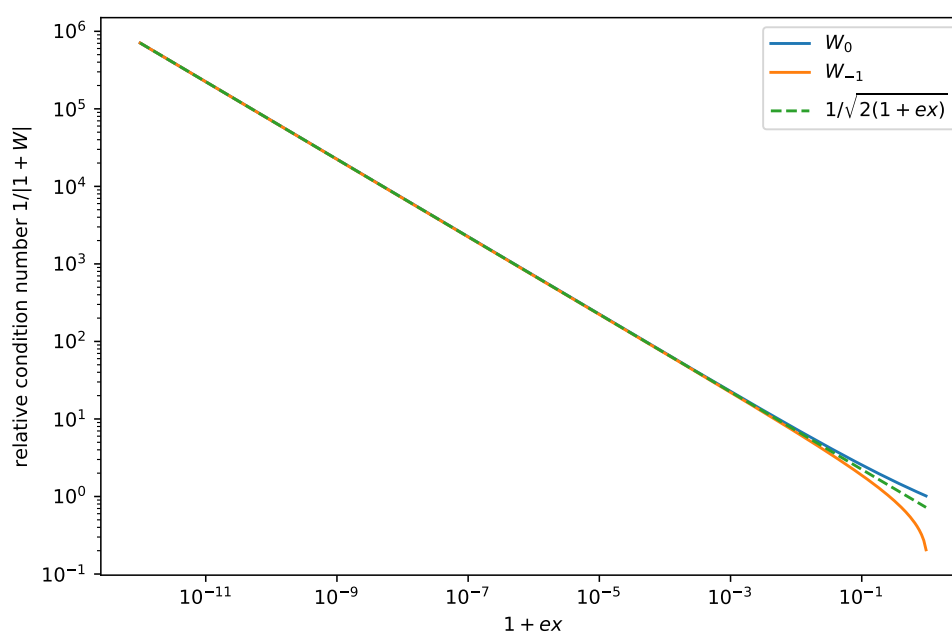


Figure 2.1: Relative conditioning of both real branches near the branch point. The square-root blow-up is the principal numerical difficulty of the problem.

Chapter 3

Solving equations with Lambert W

3.1 A disciplined reduction method

The usual goal is to transform an equation until the unknown occurs in a product of the form

$$ue^u = c.$$

Then $u = W_k(c)$. The algebra is generally easy; branch selection is the substantive step.

Proposition 3.1 (Affine exponential equation). *Let $a, c \neq 0$. Every solution of*

$$e^{ax+b} = cx + d$$

is of the form

$$x = -\frac{1}{a}W_k\left(-\frac{a}{c}\exp\left(b - \frac{ad}{c}\right)\right) - \frac{d}{c}, \quad (3.1)$$

for a branch k on which the expression is defined.

Proof. Set $y = cx + d$, so $x = (y - d)/c$. The equation becomes

$$\exp\left(\frac{a}{c}y + b - \frac{ad}{c}\right) = y.$$

Multiplying by $\exp(-ay/c)$ and then by $-a/c$ yields

$$\left(-\frac{a}{c}y\right)\exp\left(-\frac{a}{c}y\right) = -\frac{a}{c}\exp\left(b - \frac{ad}{c}\right).$$

Apply W_k , solve for y , and then for x . □

Example 3.2. The equation $2^x = x + 3$ has the form of [Theorem 3.1](#) with $a = \log 2$, $b = 0$, $c = 1$, and $d = 3$. Therefore

$$x = -\frac{1}{\log 2}W_k\left(-\log 2 \, 2^{-3}\right) - 3.$$

Because the argument lies in $(-1/e, 0)$, two real branches are available. The original equation or a graph determines whether both values satisfy the problem's constraints.

3.2 Powers times exponentials

Proposition 3.3. *Suppose $m \in \mathbb{N}$, $q \neq 0$, and a real m -th root $r^{1/m}$ has been chosen consistently. Solutions of*

$$x^m e^{qx} = r$$

that correspond to this root satisfy

$$x = \frac{m}{q} W_k \left(\frac{q}{m} r^{1/m} \right).$$

Proof. Taking the selected m -th root gives $x e^{qx/m} = r^{1/m}$. Multiply by q/m and apply W_k . For even m , the other real root may produce additional solutions and must be examined separately. $\square$

3.3 The equations $x \log x = a$ and $x^x = a$

Proposition 3.4. *For $x > 0$, the equation $x \log x = a$ is solved by*

$$x = \exp(W_k(a)) = \frac{a}{W_k(a)},$$

with the quotient interpreted by continuity at $a = 0$.

Proof. Set $u = \log x$, so $x = e^u$. Then $x \log x = u e^u = a$, hence $u = W_k(a)$, and exponentiation gives the result. The identity $e^{W(a)} = a/W(a)$ follows from the defining equation. $\square$

Theorem 3.5 (Complete real analysis of $x^x = a$). *For $x > 0$,*

$$x^x = a \iff x = \exp(W_k(\log a)).$$

Let $a_ = e^{-1/e}$. Then:*

- (a) *if $0 < a < a_*$, there is no positive real solution;*
- (b) *if $a = a_*$, there is one solution, $x = 1/e$;*
- (c) *if $a_* < a < 1$, there are two solutions, obtained from W_0 and W_{-1} ;*
- (d) *if $a \geq 1$, there is one positive solution, obtained from W_0 .*

Proof. Taking logarithms gives $x \log x = \log a$, so [Theorem 3.4](#) yields the formula. The function $g(x) = x \log x$ has derivative $1 + \log x$, hence its minimum occurs at $x = 1/e$ and equals $-1/e$. Equivalently, x^x has minimum $e^{-1/e}$. The number of solutions on either side of the minimum agrees exactly with the two real branches described in [Theorem 1.2](#). $\square$

3.4 Infinite power towers

The finite towers are defined recursively by

$$t_1 = a, \quad t_{n+1} = a^{t_n} \quad (n \geq 1),$$

where $a > 0$. If (t_n) converges to a positive limit h , continuity forces $h = a^h$. Solving this fixed-point equation gives a Lambert W formula, but the existence of a fixed point does not by itself guarantee that the iteration converges to it.

Theorem 3.6 (Exact real power-tower interval). *The sequence of finite towers (t_n) converges if and only if*

$$e^{-e} \leq a \leq e^{1/e}.$$

Throughout this interval its limit is

$$h = \begin{cases} -\frac{W_0(-\log a)}{\log a}, & a \neq 1, \\ 1, & a = 1. \end{cases} \quad (3.2)$$

For $1 < a < e^{1/e}$, the second real branch W_{-1} produces another positive solution of $h = a^h$, but that solution is repelling and is not the limit of the finite towers.

Proof. First solve the fixed-point equation. For $a \neq 1$, taking logarithms in $h = a^h$ and putting $u = -h \log a$ gives

$$ue^u = -\log a,$$

hence

$$h = -\frac{W_k(-\log a)}{\log a}.$$

For $a = 1$, every finite tower equals 1, and the value in (3.2) is also obtained by continuity.

It remains to decide when the iteration converges. Write $g(x) = a^x$, so that $t_{n+1} = g(t_n)$. At a fixed point,

$$g'(h) = h \log a = \log h = -W_k(-\log a).$$

Thus a locally attracting fixed point must satisfy $|\log h| < 1$, or $1/e < h < e$. The endpoints will be handled directly below.

Suppose first that $1 < a \leq e^{1/e}$. The function $q(h) = h^{1/h}$ increases on $[1, e]$, because

$$\frac{q'(h)}{q(h)} = \frac{1 - \log h}{h^2} \geq 0.$$

Consequently there is a unique $h \in [1, e]$ such that $a = q(h)$, and this is precisely the value obtained from W_0 . Since $a = h^{1/h} \leq h$, the first term satisfies $t_1 \leq h$. Moreover, g is increasing, $t_2 = a^a \geq a = t_1$, and $g([t_1, h]) \subseteq [t_1, h]$. Induction therefore shows that (t_n) is increasing and bounded above by h . Its limit must be a fixed point and hence equals h . This argument includes the neutral upper endpoint $a = e^{1/e}$, where $h = e$ and $g'(h) = 1$.

Now suppose that $e^{-e} \leq a < 1$, and write $a = e^{-A}$, so $0 < A \leq e$. There is a unique fixed point $h \in [1/e, 1)$, again given by W_0 . Since $a = h^{1/h} < h$ and g is decreasing,

$$t_1 < h < t_2.$$

Also every term lies in $(0, 1)$. The odd subsequence is increasing and the even subsequence is decreasing: indeed $t_3 = a^{t_2} > a = t_1$, because $t_2 < 1$, and then the monotonicity of $g \circ g$ propagates these inequalities. Both subsequences are bounded by h , so they have limits $p \leq h \leq q$ satisfying

$$p = g(q), \quad q = g(p).$$

We show that $p = q$. Otherwise $0 < p < q < 1$ would be a nontrivial two-cycle. Put $u = -\log p$ and $v = -\log q$, so $u > v > 0$. The two-cycle equations give

$$A = ue^v = ve^u,$$

and therefore $\log(u/v) = u - v$. Writing $r = u/v > 1$, we obtain

$$v = \frac{\log r}{r-1}, \quad u = \frac{r \log r}{r-1}, \quad A = \frac{r \log r}{r-1} r^{1/(r-1)}.$$

This last quantity is strictly larger than e . To see this, put $r = e^s$, $s > 0$, and let $F(s)$ be its logarithm. Direct differentiation gives

$$F'(s) = \frac{(e^s - 1)^2 - s^2 e^s}{s(e^s - 1)^2} > 0,$$

because $e^s - 1 > se^{s/2}$, which is equivalent to $2 \sinh(s/2) > s$. Since $F(s) \rightarrow 1$ as $s \downarrow 0$, we have $A > e$, contradicting $A \leq e$. Hence $p = q = h$, and the full sequence converges. This includes the neutral lower endpoint $a = e^{-e}$, where $h = 1/e$ and $g'(h) = -1$.

Finally, if $a > e^{1/e}$, then $h^{1/h} = a$ has no positive solution, because q has global maximum $q(e) = e^{1/e}$. If $0 < a < e^{-e}$, its unique fixed point lies below $1/e$ and satisfies $|g'(h)| = |\log h| > 1$. Were the distinct iterates to converge to h , the mean value theorem would eventually expand, rather than shrink, their distance from h . They cannot land on h in finitely many steps: since g is one-to-one, that would force every preceding term, including $t_1 = a$, to equal h , which is impossible for $0 < a < 1$. Thus no convergence occurs outside the stated interval.

For $1 < a < e^{1/e}$, the W_{-1} value corresponds to a second fixed point $H > e$. There $g'(H) = \log H > 1$, so it is repelling. The finite towers instead converge to the attracting W_0 fixed point in $(1, e)$. $\square$

3.5 A branch-selection checklist

For a real problem, the following short procedure prevents most mistakes.

1. Determine the transformed argument z .
2. If $z < -1/e$, no real W -value exists.
3. If $-1/e < z < 0$, write down both $W_0(z)$ and $W_{-1}(z)$.
4. Translate the ranges $[-1, 0)$ and $(-\infty, -1]$ back to the original variable.
5. Check every surviving candidate in the original equation, especially if roots or logarithms were introduced.

Part II

Series, local structure, and asymptotics

Chapter 4

Power series and Lagrange inversion

4.1 Coefficient extraction and the inversion theorem

Power series are the natural language of a locally analytic inverse. We use $[z^n]F(z)$ to denote the coefficient of z^n in a power series F . The following form of the Lagrange–Bürmann inversion theorem is sufficient for almost everything in this chapter.

Theorem 4.1 (Lagrange–Bürmann inversion). *Let ϕ be analytic near 0, with $\phi(0) \neq 0$, and let $w(z)$ be the analytic solution near the origin of*

$$w = z\phi(w), \quad w(0) = 0.$$

If F is analytic near 0, then for every integer $n \geq 1$,

$$[z^n]F(w(z)) = \frac{1}{n}[t^{n-1}]F'(t)\phi(t)^n. \quad (4.1)$$

In particular,

$$[z^n]w(z)^m = \frac{m}{n}[t^{n-m}]\phi(t)^n \quad (n \geq m \geq 1).$$

Proof. Choose small positively oriented circles on which all functions involved are analytic and on which the substitution $t = w(z)$ is one-to-one. Since $z = t/\phi(t)$, Cauchy’s coefficient formula and a change of variable give

$$\begin{aligned} [z^n]F(w(z)) &= \frac{1}{2\pi i} \int \frac{F(w(z))}{z^{n+1}} dz \\ &= \frac{1}{2\pi i} \int F(t) \left(\frac{\phi(t)^n}{t^{n+1}} - \frac{\phi(t)^{n-1}\phi'(t)}{t^n} \right) dt. \end{aligned}$$

The integrand satisfies the exact identity

$$F(t) \left(\frac{\phi(t)^n}{t^{n+1}} - \frac{\phi(t)^{n-1}\phi'(t)}{t^n} \right) = -\frac{1}{n} \frac{d}{dt} \left(\frac{F(t)\phi(t)^n}{t^n} \right) + \frac{1}{n} \frac{F'(t)\phi(t)^n}{t^n}.$$

The integral of the total derivative around the closed contour is zero. Equivalently, a derivative has residue zero. Therefore

$$[z^n]F(w(z)) = \frac{1}{2\pi i} \int \frac{F'(t)\phi(t)^n}{nt^n} dt = \frac{1}{n}[t^{n-1}]F'(t)\phi(t)^n.$$

Taking $F(t) = t^m$ proves the last assertion. □

Remark 4.2. The proof used complex integration only as a compact bookkeeping device. The identity is equally valid for formal power series, where it can be proved by coefficient manipulations. Thus it may be used before any questions about numerical convergence are considered.

4.2 The Maclaurin series of the principal branch

The relation $we^w = z$ can be written

$$w = ze^{-w}.$$

Thus $\phi(t) = e^{-t}$ in [Theorem 4.1](#).

Theorem 4.3 (Maclaurin series). *For $|z| < 1/e$,*

$$W_0(z) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} z^n = z - z^2 + \frac{3}{2}z^3 - \frac{8}{3}z^4 + \frac{125}{24}z^5 - \dots \quad (4.2)$$

The radius of convergence is exactly $1/e$. The series also converges absolutely at both endpoints $z = \pm 1/e$, and at the negative endpoint its sum is -1 .

Proof. Apply [Theorem 4.1](#) with $F(t) = t$:

$$[z^n]W_0(z) = \frac{1}{n}[t^{n-1}]e^{-nt} = \frac{1}{n} \frac{(-n)^{n-1}}{(n-1)!} = \frac{(-n)^{n-1}}{n!}.$$

Stirling's formula gives

$$\frac{n^{n-1}}{n!} \sim \frac{e^n}{\sqrt{2\pi} n^{3/2}},$$

so the root test gives radius $1/e$. This radius is also geometrically inevitable: the nearest point at which the inverse ceases to be locally analytic is the critical value $-1/e$.

At $z = \pm 1/e$, the absolute values of the terms are asymptotic to $1/(\sqrt{2\pi} n^{3/2})$, so both endpoint series converge absolutely. On $(-1/e, 0)$ the sum is the analytic inverse W_0 ; continuity up to the endpoint, or Abel's theorem, then gives the value $W_0(-1/e) = -1$. $\square$

Warning 4.4. The series (4.2) belongs to W_0 , not to W_{-1} . The lower real branch diverges to $-\infty$ as $x \rightarrow 0^-$, so it cannot possess a Taylor series at the origin. Its appropriate local expansion is logarithmic and is developed in [Section 6.3](#).

4.3 Powers and exponentials of W_0

The full strength of Lagrange inversion gives many related series at almost no additional cost.

Proposition 4.5 (Powers of W_0). *For every integer $m \geq 1$,*

$$W_0(z)^m = \sum_{n=m}^{\infty} \frac{m}{n} \frac{(-n)^{n-m}}{(n-m)!} z^n, \quad |z| < 1/e. \quad (4.3)$$

Proof. Use $F(t) = t^m$ and $\phi(t) = e^{-t}$ in [Theorem 4.1](#):

$$[z^n]W_0(z)^m = \frac{m}{n}[t^{n-m}]e^{-nt} = \frac{m}{n} \frac{(-n)^{n-m}}{(n-m)!}.$$

$\square$

Proposition 4.6 (Exponentials of W_0). *For every complex parameter a ,*

$$\exp(aW_0(z)) = 1 + \sum_{n=1}^{\infty} \frac{a(a-n)^{n-1}}{n!} z^n, \quad |z| < 1/e. \quad (4.4)$$

The formula is interpreted by continuity when $a = 0$.

Proof. For $F(t) = e^{at}$, [Theorem 4.1](#) gives

$$[z^n]e^{aW_0(z)} = \frac{a}{n} [t^{n-1}]e^{(a-n)t} = \frac{a(a-n)^{n-1}}{n!}.$$

The constant term is 1. □

Setting $a = -1$, for example, gives a series for $e^{-W_0(z)} = W_0(z)/z$. Differentiating (4.2) gives series for W'_0 and, through $zW'_0 = W_0/(1+W_0)$, for rational expressions in W_0 .

4.4 Taylor series about an arbitrary regular point

The origin is not privileged. Let $w_* \neq -1$, put $z_* = w_*e^{w_*}$, and choose the local inverse branch satisfying $W(z_*) = w_*$. Taylor's theorem and the derivative formulas of [Chapter 2](#) give

$$\begin{aligned} W(z_* + \Delta) &= w_* + \frac{e^{-w_*}}{1+w_*} \Delta - \frac{e^{-2w_*}(w_*+2)}{2(1+w_*)^3} \Delta^2 \\ &\quad + \frac{e^{-3w_*}(2w_*^2+8w_*+9)}{6(1+w_*)^5} \Delta^3 + \mathcal{O}(\Delta^4). \end{aligned} \quad (4.5)$$

This expansion is valid on either real branch. The coefficients become large as $w_* \rightarrow -1$, signaling that an ordinary Taylor series is the wrong local coordinate at the branch point. Formula (4.5) is also the basis of the Schröder-type corrections in [Section 9.6](#).

Theorem 4.7 (Local inverse series on both real branches). *Let $w_* \in (-\infty, -1) \cup (-1, \infty)$, and $z_* = w_*e^{w_*}$. Then there is a disk centered at z_* on which a unique analytic inverse branch satisfies $W(z_*) = w_*$, and its Taylor coefficients are $W^{(n)}(z_*)/n!$, with $W^{(n)}$ given by (2.4).*

Proof. The derivative of $f(w) = we^w$ at w_* is $e^{w_*}(1+w_*) \neq 0$. The analytic inverse function theorem therefore produces a unique analytic local inverse. Taylor's theorem supplies the series, and [Theorem 2.3](#) supplies every coefficient. □

4.5 The tree function

The sign change

$$T(z) = -W_0(-z)$$

produces the classical tree function. From (4.2),

$$T(z) = \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!} z^n, \quad |z| < 1/e, \quad (4.6)$$

and the defining relation is

$$T(z)e^{-T(z)} = z, \quad T(z) = ze^{T(z)}.$$

The coefficient n^{n-1} counts rooted labeled trees on n vertices. A complete combinatorial explanation is given in [Chapter 11](#).

Chapter 5

The square-root branch point

5.1 Why ordinary Taylor series fail

At $w = -1$, the derivative of $f(w) = we^w$ vanishes, while $f''(-1) = e^{-1} \neq 0$. Thus the graph has a nondegenerate quadratic turning point. Inverting a quadratic relation produces a square root, so the correct local series is a Puiseux series, a power series in a square-root variable.

Write

$$w = -1 + u, \quad p = \sqrt{2(1 + ex)}.$$

Then

$$\frac{p^2}{2} = 1 + (-1 + u)e^u = \frac{u^2}{2} + \frac{u^3}{3} + \frac{u^4}{8} + \frac{u^5}{30} + \cdots. \quad (5.1)$$

The principal square root gives $p \geq 0$ for real $x \geq -1/e$.

5.2 An all-order coefficient formula

Define the analytic function

$$\Phi(u) = \left(\frac{2\{1 + (-1 + u)e^u\}}{u^2} \right)^{1/2}, \quad \Phi(0) = 1. \quad (5.2)$$

Equation (5.1) is $p = u\Phi(u)$, or $u = p/\Phi(u)$. Lagrange inversion applies once more.

Theorem 5.1 (Puiseux expansion at $-1/e$). *There is a convergent power series*

$$U(p) = \sum_{n=1}^{\infty} c_n p^n$$

near $p = 0$, where

$$c_n = \frac{1}{n} [t^{n-1}] \Phi(t)^{-n}, \quad (5.3)$$

with Φ defined by (5.2). The two real branches are obtained from the same series by

$$W_0(x) = -1 + U(p), \quad (5.4)$$

$$W_{-1}(x) = -1 + U(-p), \quad (5.5)$$

where $p = \sqrt{2(1+ex)}$. The first terms are

$$\begin{aligned} U(p) = p - \frac{1}{3}p^2 + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \frac{769}{17280}p^5 - \frac{221}{8505}p^6 \\ + \frac{680863}{43545600}p^7 - \frac{1963}{204120}p^8 + \frac{226287557}{37623398400}p^9 - \frac{5776369}{1515591000}p^{10} + \mathcal{O}(p^{11}). \end{aligned} \quad (5.6)$$

Proof. The quotient inside Φ has a removable singularity at the origin and has value 1 there, as follows from (5.1). Hence Φ is analytic and nonzero near 0. The analytic inverse function theorem gives a unique analytic solution $u = U(p)$ of $p = u\Phi(u)$, with $U(0) = 0$ and $U'(0) = 1$. Applying Theorem 4.1 to $u = p\Phi(u)^{-1}$ yields (5.3). Substitution and coefficient comparison give the displayed rational numbers.

For real $x > -1/e$, the solution with $u > 0$ belongs to the range $(-1, \infty)$ and is therefore W_0 ; this is $U(p)$. The solution with $u < 0$ is obtained by choosing the other sign of the square root, hence by replacing p by $-p$. On the real interval where it exists, this is W_{-1} . This proves both branch formulas. $\square$

Corollary 5.2 (Leading branch-point behavior). *As $x \downarrow -1/e$, with $p = \sqrt{2(1+ex)}$,*

$$\begin{aligned} W_0(x) &= -1 + p - \frac{1}{3}p^2 + \mathcal{O}(p^3), \\ W_{-1}(x) &= -1 - p - \frac{1}{3}p^2 + \mathcal{O}(p^3). \end{aligned}$$

Consequently,

$$W_0(x) - W_{-1}(x) = 2p + \frac{11}{36}p^3 + \mathcal{O}(p^5),$$

and

$$W_0(x) + W_{-1}(x) = -2 - \frac{2}{3}p^2 - \frac{43}{270}p^4 + \mathcal{O}(p^6).$$

Proof. Insert p and $-p$ into (5.6). Odd powers change sign and even powers do not. Addition and subtraction give the two final formulas. $\square$

5.3 Derivatives and conditioning revisited

Differentiating the leading terms, or using Theorem 2.1, gives

$$\begin{aligned} W'_0(x) &\sim \frac{e}{p}, \\ W'_{-1}(x) &\sim -\frac{e}{p}, \end{aligned}$$

where $p = \sqrt{2(1+ex)}$. The two graphs therefore meet with vertical opposite tangents. No finite-order polynomial in $x + 1/e$ can represent this behavior; the square-root coordinate is essential.

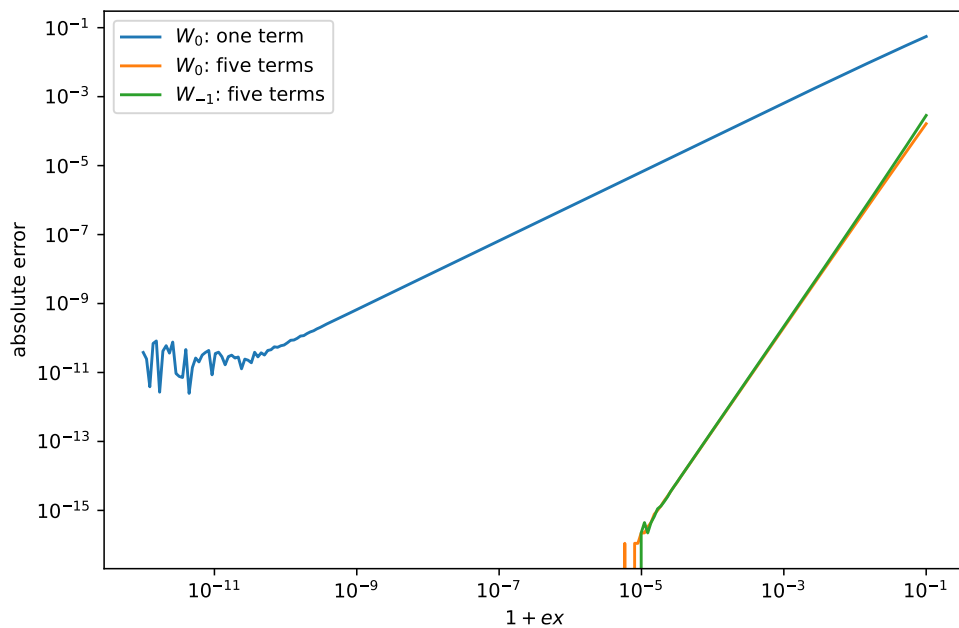


Figure 5.1: Errors of truncated Puiseux expansions as the branch point is approached. The same coefficient table serves both real branches.

5.4 A practical branch-point coordinate

The exact identity

$$x = \frac{p^2/2 - 1}{e}$$

means that no further inversion is needed once p is known. For numerical work, the expression $1 + ex$ can suffer cancellation when x is extremely close to $-1/e$. If the input itself is supplied as $x = -1/e + \delta$, it is preferable to compute $p = \sqrt{2e\delta}$ directly. Likewise, if $x = -e^{-1-u}$, then

$$p = \sqrt{2(1 - e^{-u})},$$

which can be evaluated accurately with an `expm1`-type routine.

Chapter 6

Logarithmic asymptotic expansions

6.1 The universal asymptotic polynomials

Large values of W are found by taking logarithms of the defining relation:

$$W + \operatorname{Log} W = \operatorname{Log} z, \quad (6.1)$$

with compatible logarithm branches. A family of polynomials organizes the successive corrections.

Definition 6.1 (Asymptotic polynomials). Let $P_n(\ell)$ be determined recursively by

$$P_n(\ell) = -[t^n] \log \left(1 - \ell t + \sum_{j=1}^{n-1} P_j(\ell) t^{j+1} \right), \quad n \geq 1. \quad (6.2)$$

Then

$$\begin{aligned} P_1(\ell) &= \ell, \\ P_2(\ell) &= \frac{\ell(\ell-2)}{2}, \\ P_3(\ell) &= \frac{\ell(2\ell^2-9\ell+6)}{6}, \\ P_4(\ell) &= \frac{\ell(3\ell^3-22\ell^2+36\ell-12)}{12}, \\ P_5(\ell) &= \frac{\ell(12\ell^4-125\ell^3+350\ell^2-300\ell+60)}{60}, \\ P_6(\ell) &= \frac{\ell(20\ell^5-274\ell^4+1125\ell^3-1700\ell^2+900\ell-120)}{120}. \end{aligned}$$

The recurrence is constructive: to find P_n , only the earlier polynomials are required. There is also a closed finite sum. Let $\begin{bmatrix} n \\ j \end{bmatrix}$ denote an unsigned Stirling number of the first kind.

Proposition 6.2 (Stirling-number formula). For $n \geq 1$,

$$P_n(\ell) = \sum_{m=1}^n (-1)^{n-m} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} \frac{\ell^m}{m!}. \quad (6.3)$$

Proof. The bivariate formal expansion

$$\sum_{r=0}^{\infty} \sum_{m=1}^{\infty} (-1)^r \begin{bmatrix} r+m \\ r+1 \end{bmatrix} \frac{\ell^m}{m!} t^{r+m} \quad (6.4)$$

solves, coefficient by coefficient, the equation

$$R(t) + \log(1 - \ell t + tR(t)) = 0, \quad R(t) = \sum_{n \geq 1} P_n(\ell) t^n.$$

Indeed, the standard generating identity

$$\frac{(-\log(1-s))^{r+1}}{(r+1)!} = \sum_{n \geq r+1} \begin{bmatrix} n \\ r+1 \end{bmatrix} \frac{s^n}{n!}$$

reduces the coefficient verification to the power-series expansion of the logarithm. The coefficient of t^n in (6.4) is obtained by setting $r = n - m$, and is exactly (6.3). Uniqueness of the coefficient recursion (6.2) completes the proof. $\square$

6.2 Large positive arguments on W_0

Put

$$L = \log x, \quad \ell = \log L.$$

The first guess $W_0(x) \approx L$ is improved by observing that $W + \log W = L$, so the next guess is $L - \ell$.

Theorem 6.3 (All-order expansion of W_0). *For every fixed integer $N \geq 0$, as $x \rightarrow +\infty$,*

$$W_0(x) = L - \ell + \sum_{n=1}^N \frac{P_n(\ell)}{L^n} + \mathcal{O}\left(\frac{\ell^{N+1}}{L^{N+1}}\right), \quad (6.5)$$

where an empty sum is understood when $N = 0$. In particular,

$$\begin{aligned} W_0(x) = L - \ell + \frac{\ell}{L} + \frac{\ell(\ell-2)}{2L^2} + \frac{\ell(2\ell^2-9\ell+6)}{6L^3} \\ + \frac{\ell(3\ell^3-22\ell^2+36\ell-12)}{12L^4} + \dots \end{aligned} \quad (6.6)$$

Proof. Set $t = 1/L$ and seek

$$W = L - \ell + R, \quad R = \sum_{n \geq 1} P_n(\ell) t^n.$$

Then

$$\log W = \ell + \log(1 - \ell t + tR),$$

so $W + \log W - L = 0$ becomes

$$R + \log(1 - \ell t + tR) = 0.$$

Equating powers of t produces exactly (6.2), proving the formal expansion.

For the asymptotic remainder, truncate after N terms and call the result S_N . Direct substitution into $F(w) = w + \log w - L$ gives

$$F(S_N) = \mathcal{O}\left(\frac{\ell^{N+1}}{L^{N+1}}\right).$$

For large x , both S_N and the exact root lie near L , and $F'(w) = 1 + 1/w$ is bounded away from zero. The mean value theorem therefore gives

$$W_0(x) - S_N = -\frac{F(S_N)}{F'(\xi)} = \mathcal{O}\left(\frac{\ell^{N+1}}{L^{N+1}}\right)$$

for some ξ between S_N and $W_0(x)$. □

6.3 The lower branch near the origin

Let $x \rightarrow 0^-$, and define

$$T = \log \frac{1}{-x}, \quad \ell = \log T.$$

Writing $W_{-1}(x) = -u$, the defining equation becomes

$$ue^{-u} = -x, \quad u - \log u = T.$$

The sign difference before $\log u$ produces alternating signs in the correction polynomials.

Theorem 6.4 (All-order expansion of W_{-1}). *For every fixed integer $N \geq 0$, as $x \rightarrow 0^-$,*

$$W_{-1}(x) = -T - \ell - \sum_{n=1}^N (-1)^{n-1} \frac{P_n(\ell)}{T^n} + \mathcal{O}\left(\frac{\ell^{N+1}}{T^{N+1}}\right). \quad (6.7)$$

Thus

$$\begin{aligned} W_{-1}(x) = & -T - \ell - \frac{\ell}{T} + \frac{\ell(\ell-2)}{2T^2} - \frac{\ell(2\ell^2-9\ell+6)}{6T^3} \\ & + \frac{\ell(3\ell^3-22\ell^2+36\ell-12)}{12T^4} + \dots \end{aligned} \quad (6.8)$$

Proof. Seek

$$u = T + \ell + \sum_{n \geq 1} \frac{Q_n(\ell)}{T^n}.$$

Substitution into $u - \log u = T$, with $t = 1/T$, gives

$$Q - \log(1 + \ell t + tQ) = 0, \quad Q = \sum_{n \geq 1} Q_n(\ell) t^n.$$

Replacing t by $-t$ in the defining equation for the polynomials P_n shows coefficient by coefficient that

$$Q_n(\ell) = (-1)^{n-1} P_n(\ell).$$

Since $W_{-1} = -u$, this gives the formal series (6.7). The remainder proof is the same mean-value argument as in [Theorem 6.3](#), now applied to $G(u) = u - \log u - T$. For large T , $G'(u) = 1 - 1/u$ is bounded away from zero. □

The leading two terms are often written directly in terms of x :

$$W_{-1}(x) = \log(-x) - \log(-\log(-x)) + o(1). \quad (6.9)$$

This compact form is useful but converges slowly unless $-x$ is extremely small.

6.4 Complex branches

For a chosen branch index $k \in \mathbb{Z}$, set

$$L_1 = \operatorname{Log} z + 2\pi i k, \quad L_2 = \operatorname{Log} L_1,$$

with logarithms chosen consistently in the region under consideration. The same algebra used above gives the branch-uniform formal expansion

$$W_k(z) \sim L_1 - L_2 + \sum_{n=1}^{\infty} \frac{P_n(L_2)}{L_1^n}. \quad (6.10)$$

For fixed k , it applies as $z \rightarrow \infty$ away from the relevant cuts. It also describes $W_k(z)$ as $z \rightarrow 0$ when $k \neq 0$, because then $|L_1| \rightarrow \infty$. This explains the logarithmic branch point of every nonprincipal branch at the origin.

Derivation. The logarithmic equation is $W + \operatorname{Log} W = L_1$. Put $W = L_1 - L_2 + R$ and $t = 1/L_1$. Exactly as in the proof of [Theorem 6.3](#), one obtains

$$R + \operatorname{Log}(1 - L_2 t + t R) = 0,$$

whose unique formal solution is $R = \sum_{n \geq 1} P_n(L_2) t^n$. Standard analytic estimates make the expansion asymptotic on closed subsectors that avoid branch cuts and turning points; the algebraic coefficient formula itself is independent of the sector. $\square$

Remark 6.5. The real lower-branch expansion (6.7) can be obtained from the complex formula by approaching the negative real axis with the appropriate logarithm determinations. The direct real derivation above is preferable for students because it avoids a delicate cancellation of imaginary parts.

Chapter 7

Integral, contour, continued-fraction, and limit representations

7.1 The master substitution

Every integral involving a single branch of W can be transferred to the w -plane by

$$x = we^w, \quad dx = e^w(1+w) dw. \quad (7.1)$$

The substitution is valid on any interval or contour on which one branch is followed continuously and the turning point $w = -1$ is handled as an endpoint if it occurs.

Theorem 7.1 (Integration by the inverse variable). *Let F be continuous on the image of a branch segment. Then*

$$\int F(W(x)) dx = \int F(w)e^w(1+w) dw, \quad w = W(x). \quad (7.2)$$

More generally,

$$\int x^a W(x)^b dx = \int w^{a+b}(1+w)e^{(a+1)w} dw, \quad (7.3)$$

whenever the powers and branches are chosen consistently.

Proof. Both formulas are the ordinary change-of-variables theorem applied to (7.1). For the second, use $x^a = (we^w)^a$. $\square$

When $a+b$ is a nonnegative integer, the right side of (7.3) is elementary by repeated integration by parts. For general exponents it can be expressed through incomplete gamma functions. Thus many apparently exotic integrals involving W reduce to standard exponential moments.

7.2 Useful antiderivatives

Several formulas are worth memorizing.

Proposition 7.2 (Elementary antiderivatives). *On any regular branch interval,*

$$\int W(x) dx = x \left(W(x) - 1 + \frac{1}{W(x)} \right) + C, \quad (7.4)$$

$$\int \frac{W(x)}{x} dx = W(x) + \frac{1}{2} W(x)^2 + C, \quad (7.5)$$

$$\int \frac{dx}{1 + W(x)} = \frac{x}{W(x)} + C = e^{W(x)} + C, \quad (7.6)$$

$$\int \frac{dx}{x(1 + W(x))} = \log |W(x)| + C. \quad (7.7)$$

At removable points such as $x = 0$ on W_0 , the right sides are understood by continuity.

Proof. For (7.4), substitute (7.1):

$$\int w e^w (1 + w) dw = e^w (w^2 - w + 1) + C = x \left(w - 1 + \frac{1}{w} \right) + C.$$

For (7.5), the derivative identity gives

$$\frac{dx}{x} = \frac{1 + w}{w} dw, \quad \frac{W(x)}{x} dx = (1 + w) dw.$$

The last two formulas follow respectively from $dx/(1+w) = e^w dw$ and $W'/W = 1/[x(1+W)]$. $\square$

Example 7.3 (Definite integrals over the two negative branches). Using the antiderivative $e^w(w^2 - w + 1)$,

$$\begin{aligned} \int_{-1/e}^0 W_0(x) dx &= 1 - \frac{3}{e}, \\ \int_{-1/e}^0 W_{-1}(x) dx &= -\frac{3}{e}. \end{aligned}$$

For the lower branch, w runs from -1 to $-\infty$, and $e^w(w^2 - w + 1) \rightarrow 0$. The logarithmic divergence of $W_{-1}(x)$ at zero is therefore integrable.

Example 7.4 (An omega-constant integral). Let $\Omega = W_0(1)$. Since the continuous limit of the antiderivative (7.4) at $x = 0$ is 1,

$$\int_0^1 W_0(x) dx = \Omega - 2 + \frac{1}{\Omega}.$$

7.3 A contour formula that isolates any simple branch value

The argument principle gives a representation that treats all branches on the same footing.

Theorem 7.5 (Root-isolating contour representation). *Fix $z \in \mathbb{C}$, and let Γ be a positively oriented simple closed contour containing exactly one simple solution $w_* = W_k(z)$ of $we^w = z$, with no solution on Γ . Then*

$$W_k(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{w(1+w)e^w}{we^w - z} dw. \quad (7.8)$$

More generally, replacing w in the numerator by an analytic function $F(w)$ returns $F(W_k(z))$.

Proof. Let $h(w) = we^w - z$. At a simple zero w_* , $h'(w_*) = (1 + w_*)e^{w_*} \neq 0$. The integrand in (7.8) has residue

$$\frac{w_* h'(w_*)}{h'(w_*)} = w_*.$$

There are no other poles inside Γ . The residue theorem gives the formula. The proof with $F(w)$ is identical. $\square$

At $z = -1/e$, the two real roots coalesce into a double root and the hypothesis of simplicity fails. The value -1 is recovered by taking a limit from either side of the branch point.

7.4 A real Fourier-logarithm integral for W_0

Corcino, Corcino, and Mező derived an unusual integral representation in which the Maclaurin coefficients are selected by Fourier orthogonality [4].

Theorem 7.6 (Corcino–Corcino–Mező integral). *For real $-1/e \leq x \leq e$,*

$$W_0(x) = \frac{1}{\pi} \operatorname{Re} \int_0^\pi \log \left(\frac{\exp(e^{it}) - xe^{-it}}{\exp(e^{it}) - xe^{it}} \right) dt, \quad (7.9)$$

where the logarithm is followed continuously along the integration path. At the endpoints the integral is interpreted as an improper limit.

Proof. First suppose $|x| < 1/e$. Since $|xe^{\pm it - e^{it}}| \leq e|x| < 1$, both logarithms may be expanded uniformly. Dividing numerator and denominator by $\exp(e^{it})$ gives

$$\begin{aligned} & \log(1 - xe^{-it - e^{it}}) - \log(1 - xe^{it - e^{it}}) \\ &= \sum_{n=1}^{\infty} \frac{x^n}{n} e^{-ne^{it}} (e^{int} - e^{-int}). \end{aligned}$$

Expand once more,

$$e^{-ne^{it}} = \sum_{j=0}^{\infty} \frac{(-n)^j}{j!} e^{ij t}.$$

The real part of the integral over $[0, \pi]$ vanishes for every Fourier mode except the constant mode. In the difference above, the only constant mode occurs in the second term when $j = n$. Therefore the coefficient of x^n in the right side of (7.9) is

$$\frac{1}{\pi n} \left(-\pi \frac{(-n)^n}{n!} \right) = \frac{(-n)^{n-1}}{n!}.$$

The result is consequently the Maclaurin series of W_0 .

For real $-1/e < x < e$, neither numerator nor denominator in the logarithm vanishes for $0 < t < \pi$. Indeed, a zero would require $x = \exp(e^{it} \mp it)$ to be real, but the imaginary parts $\sin t \mp t$ attain the required multiples of π only at the endpoints. Hence the real part of the integral is real-analytic in x on the connected interval $(-1/e, e)$. It agrees with W_0 on the nonempty subinterval $(-1/e, 1/e)$, so uniqueness of analytic continuation gives equality throughout. At $x = -1/e$ and $x = e$, the endpoint logarithmic singularities are integrable, and continuity gives the endpoint values. $\square$

Remark 7.7. Formula (7.9) is a representation of the principal branch. It does not automatically produce W_{-1} , because its proof starts from the Maclaurin series at the origin, where the lower branch is not analytic.

7.5 Continued fractions

A continued fraction may encode more information than a polynomial of the same formal order, but it can also introduce artificial poles. Corcino, Corcino, and Mező obtained the regular C-fraction beginning

$$W_0(x) = \frac{x}{1 + \frac{x}{1 + \frac{\frac{1}{2}x}{1 + \frac{\frac{5}{6}x}{1 + \frac{\frac{17}{30}x}{1 + \ddots}}}}}. \quad (7.10)$$

It is understood first as a formal continued fraction whose convergents match successively more coefficients of (4.2). For example,

$$x, \quad \frac{x}{1+x}, \quad \frac{x(2+x)}{2+3x}$$

are the first three convergents. Expanding them at the origin gives x , then $x - x^2 + x^3 - \dots$, and then agreement through additional Maclaurin coefficients.

The general C-fraction coefficients can be expressed by Hankel determinants of the sequence $a_0 = 0$, $a_n = (-n)^{n-1}/n!$. Their existence theory and numerical behavior are developed in [4]. One should not infer uniform convergence on the whole real branch merely from formal coefficient agreement; the poles and the domain of each convergent must be checked.

7.6 Nested exponentials and logarithms

The defining relation itself produces fixed-point representations.

For $0 < x < e$, the equation $w = xe^{-w}$ has fixed-point derivative $-w$ at $w = W_0(x)$, whose magnitude is less than one. Thus, for starts in a suitable neighborhood,

$$W_0(x) = x \exp(-x \exp(-x \exp(\dots))). \quad (7.11)$$

For $x > e$, the rearrangement $w = \log x - \log w$ has derivative $-1/w$, whose magnitude is less than one at the fixed point. It gives a locally convergent nested-logarithm representation.

The lower real branch has an especially clean global version. Write $x = -e^{-T}$, where $T > 1$, and let $u = -W_{-1}(x) > 1$. Then

$$u = T + \log u. \quad (7.12)$$

Theorem 7.8 (Monotone nested logarithms for W_{-1}). *Let $T > 1$, define $u_0 = T$, and*

$$u_{n+1} = T + \log u_n.$$

Then u_n increases to the unique solution of (7.12), and

$$W_{-1}(-e^{-T}) = -\lim_{n \rightarrow \infty} (T + \log(T + \log(T + \dots))). \quad (7.13)$$

Proof. The function $g(u) = T + \log u$ is increasing on $(0, \infty)$. Its fixed point u_* is unique on $(1, \infty)$, because $u - \log u$ is strictly increasing there. Since $u_* > T = u_0$, induction gives $u_n < u_*$, and $u_{n+1} - u_n = \log(u_n/u_{n-1}) > 0$. Thus u_n increases and is bounded, so it has a limit. Passing to the limit in the recurrence identifies that limit as u_* . $\square$

7.7 The Wright omega function

For positive real x , one may avoid forming a huge exponential by working with

$$\omega(y) = W_0(e^y).$$

It is the unique positive solution of

$$\omega + \log \omega = y, \tag{7.14}$$

and

$$\omega'(y) = \frac{\omega(y)}{1 + \omega(y)}.$$

The relation $W_0(x) = \omega(\log x)$ is numerically useful when x is represented through its logarithm. Complex versions of ω absorb the integer multiples of $2\pi i$ that otherwise occur in the branch index of W_k ; see [6, 5].

Part III

Bounds, approximations, and computation

Chapter 8

Inequalities and certified enclosures

8.1 Elementary global bounds for the principal branch

The geometry of W_0 immediately gives useful inequalities that are often more informative than a decimal approximation.

Theorem 8.1 (Bounds for W_0 on the nonnegative axis). *For $x \geq 0$,*

$$\frac{x}{1+x} \leq W_0(x) \leq \log(1+x) \leq x. \quad (8.1)$$

All three inequalities are strict when $x > 0$, except that the displayed functions have the common value zero at the origin.

Proof. Write $x = we^w$, where $w = W_0(x) \geq 0$. Define

$$H(w) = 1 + we^w - e^w = 1 + (w-1)e^w.$$

Then $H(0) = 0$ and $H'(w) = we^w \geq 0$, so $H(w) \geq 0$. This gives $e^w \leq 1+x$, hence $w \leq \log(1+x)$. It also gives

$$w(1+x) - x = w\{1 + (w-1)e^w\} = wH(w) \geq 0,$$

which is equivalent to $x/(1+x) \leq w$. Finally, $\log(1+x) \leq x$ is the standard logarithm inequality, proved for example by integrating $1/(1+t) \leq 1$ from 0 to x . $\square$

The lower bound $x/(1+x)$ is exactly the $[1/1]$ Padé approximant of W_0 at zero. Thus a local rational approximation unexpectedly becomes a global one-sided bound on the positive axis.

Theorem 8.2 (Bounds for W_0 on its negative real interval). *For $-1/e \leq x \leq 0$,*

$$ex \leq W_0(x) \leq x. \quad (8.2)$$

Proof. The function W_0 is concave on the whole interval. A concave graph lies below every tangent line and above every chord. The tangent at the origin is $y = x$, because $W_0(0) = 0$ and $W_0'(0) = 1$, giving the upper bound. The chord joining $(-1/e, -1)$ and $(0, 0)$ is $y = ex$, giving the lower bound. $\square$

8.2 Logarithmic bounds for large positive arguments

The first two terms of the asymptotic expansion can be converted into a rigorous enclosure without assuming that x is astronomically large.

Theorem 8.3 (Two-logarithm enclosure for W_0). *Let $x > e$, $L = \log x$, $\ell = \log L$, and $A = L - \ell$. Then*

$$A < W_0(x) < A + \frac{\ell}{A}. \quad (8.3)$$

Proof. Let $w = W_0(x)$. Since $w + \log w = L$, and $w < L$, we have $\log w < \log L = \ell$, hence $w = L - \log w > A$.

Using $w > A$,

$$w = L - \log w = A + \log \frac{L}{w} < A + \log \frac{L}{A}.$$

Now $A = L(1 - \ell/L)$, so

$$\log \frac{L}{A} = -\log \left(1 - \frac{\ell}{L}\right) \leq \frac{\ell/L}{1 - \ell/L} = \frac{\ell}{A},$$

where $-\log(1 - u) \leq u/(1 - u)$ for $0 < u < 1$. This proves the upper bound. $\square$

Iterating $w = L - \log w$ from the two sides yields increasingly tight nested logarithmic enclosures. The asymptotic series of [Theorem 6.3](#) explains the correction ℓ/A and supplies much higher accuracy when desired.

8.3 Global logarithmic bounds for the lower branch

Let

$$T = \log \frac{1}{-x} > 1, \quad u = -W_{-1}(x) > 1.$$

Then $u = T + \log u$.

Theorem 8.4 (Global two-sided bounds for W_{-1}). *For $-1/e < x < 0$,*

$$-T - \frac{T \log T}{T - 1} \leq W_{-1}(x) < -T - \log T. \quad (8.4)$$

Proof. Because $u > T$, the fixed-point equation gives

$$u = T + \log u > T + \log T,$$

which becomes the right-hand inequality after multiplication by -1 .

Set

$$U = T + \frac{T \log T}{T - 1}.$$

We show that $T + \log U \leq U$. Since

$$U = T \left(1 + \frac{\log T}{T - 1}\right),$$

we have

$$\begin{aligned}\log U &= \log T + \log \left(1 + \frac{\log T}{T-1}\right) \\ &\leq \log T + \frac{\log T}{T-1} = \frac{T \log T}{T-1} = U - T.\end{aligned}$$

The increasing function $g(v) = T + \log v$ therefore maps the interval $[T, U]$ into itself. Its unique fixed point u lies in that interval, so $u \leq U$. Negating gives the left-hand inequality. $\square$

The upper enclosure deteriorates when $T \downarrow 1$, precisely where the square-root expansion is the correct tool. The two methods complement each other rather than compete.

Corollary 8.5 (Certified nested-logarithm enclosure). *With $T > 1$, let*

$$a_0 = T, \quad b_0 = T + \frac{T \log T}{T-1},$$

and define

$$a_{n+1} = T + \log a_n, \quad b_{n+1} = T + \log b_n.$$

Then

$$a_n \uparrow u = -W_{-1}(-e^{-T}), \quad b_n \downarrow u,$$

and

$$0 \leq b_n - a_n \leq T^{-n}(b_0 - a_0). \quad (8.5)$$

Proof. The proof of [Theorem 8.4](#) shows $a_0 \leq u \leq b_0$, and the increasing map $g(v) = T + \log v$ preserves this order. Moreover, $g(a_0) > a_0$ and $g(b_0) \leq b_0$, so the two sequences are monotone. On $[T, b_0]$, $g'(v) = 1/v \leq 1/T$. The mean value theorem therefore gives $b_{n+1} - a_{n+1} \leq T^{-1}(b_n - a_n)$, and induction proves the error bound. $\square$

8.4 Branch-point bounds for both real branches

A particularly convenient parametrization is

$$x = -e^{-1-q}, \quad q > 0, \quad t = \sqrt{2q}. \quad (8.6)$$

It maps $q = 0$ to the branch point and $q \rightarrow \infty$ to $x \rightarrow 0^-$.

Theorem 8.6 (Chatzigeorgiou bounds for W_{-1}). *For $q > 0$,*

$$-1 - \sqrt{2q} - q < W_{-1}(-e^{-1-q}) < -1 - \sqrt{2q} - \frac{2}{3}q. \quad (8.7)$$

Proof. Write $W_{-1}(-e^{-1-q}) = -1 - s$, where $s > 0$. The defining relation reduces to

$$(1+s)e^{-s} = e^{-q}, \quad h(s) := s - \log(1+s) = q.$$

The function h is strictly increasing for $s > 0$. Put $t = \sqrt{2q}$, so $q = t^2/2$. We will prove

$$h\left(t + \frac{t^2}{3}\right) < \frac{t^2}{2} < h\left(t + \frac{t^2}{2}\right).$$

For the right inequality,

$$h\left(t + \frac{t^2}{2}\right) > \frac{t^2}{2} \iff t > \log\left(1 + t + \frac{t^2}{2}\right),$$

which follows from $e^t > 1 + t + t^2/2$.

For the left inequality, define

$$F(t) = \log\left(1 + t + \frac{t^2}{3}\right) - t + \frac{t^2}{6}.$$

A direct calculation gives

$$F'(t) = \frac{t^3}{3(t^2 + 3t + 3)} > 0, \quad F(0) = 0.$$

Thus $F(t) > 0$, which is exactly the desired left inequality. Since h is increasing,

$$t + \frac{t^2}{3} < s < t + \frac{t^2}{2}.$$

Using $t^2 = 2q$ and $W_{-1} = -1 - s$ yields (8.7). This proof follows the bounds introduced in [3]. $\square$

The same parametrization yields a simple companion result for the principal branch.

Theorem 8.7 (Companion branch-point bounds for W_0). *For $q > 0$,*

$$-1 + \sqrt{q^2 + 2q} - q < W_0(-e^{-1-q}) < -1 + \min\{1, \sqrt{2q}\}. \quad (8.8)$$

Proof. Write $W_0(-e^{-1-q}) = -1 + s$, where $0 < s < 1$. The defining relation becomes

$$(1 - s)e^s = e^{-q}, \quad q = -s - \log(1 - s) = \sum_{n=2}^{\infty} \frac{s^n}{n}.$$

The first term gives $q > s^2/2$, hence $s < \sqrt{2q}$; also $s < 1$. For the other direction,

$$q < \frac{1}{2} \sum_{n=2}^{\infty} s^n = \frac{s^2}{2(1 - s)}.$$

Thus $s^2 + 2qs - 2q > 0$. The positive root of the corresponding quadratic is $\sqrt{q^2 + 2q} - q$, so $s > \sqrt{q^2 + 2q} - q$. Substituting $W_0 = -1 + s$ proves the result. $\square$

8.5 Turning residuals into rigorous error bounds

A small residual is not automatically a small error near a poorly conditioned root. The derivative must also be taken into account.

Theorem 8.8 (Residual enclosure in the original variable). *Let $r = W_k(x)$ be a simple real branch value, and suppose y and r lie in a closed interval I that does not contain -1 . Put*

$$m = \min_{w \in I} |e^w(1 + w)| > 0.$$

Then

$$|y - r| \leq \frac{|ye^y - x|}{m}. \quad (8.9)$$

Proof. Apply the mean value theorem to $f(w) = we^w$:

$$ye^y - re^r = f'(\xi)(y - r)$$

for some ξ between y and r . Since $re^r = x$ and $|f'(\xi)| \geq m$, the inequality follows. $\square$

A logarithmic residual is often safer against overflow and underflow. On an interval not crossing 0 or -1 , let

$$R_{\log}(y) = y + \log |y| - \log |x|.$$

If $m_{\log} = \min_I |1 + 1/w| > 0$, the same proof gives

$$|y - W_k(x)| \leq \frac{|R_{\log}(y)|}{m_{\log}}. \quad (8.10)$$

The blow-up of these bounds as the interval approaches -1 is not a defect: it faithfully records the intrinsic condition number from [Theorem 2.4](#).

Chapter 9

Analytical approximations

9.1 Approximation is a regional problem

No single elementary formula naturally reflects all four important geometries: analyticity at the origin of W_0 , square-root behavior at $-1/e$, logarithmic growth of W_0 at $+\infty$, and logarithmic divergence of W_{-1} at 0^- . A reliable approximation strategy therefore begins by identifying the region.

Table 9.1: Natural approximations in the different real-branch regions.

region	branch	preferred representation
near $x = 0$	W_0	Maclaurin or Padé expansion
near $x = -1/e$	both	Puiseux series in $p = \sqrt{2(1+ex)}$
large $x > 0$	W_0	logarithmic asymptotic expansion
small $-x > 0$	W_{-1}	lower-branch logarithmic expansion
remaining compact intervals	either	simple seed plus inverse iteration

This regional view is also the organizing principle of most high-quality software implementations and of the approximation literature in the attached archive [2, 10, 8, 9, 12].

9.2 Truncated Maclaurin series with an explicit error bound

Let

$$M_N(z) = \sum_{n=1}^N \frac{(-n)^{n-1}}{n!} z^n.$$

The following bound is simple rather than sharp, but it is completely explicit.

Theorem 9.1 (Maclaurin tail bound). *If $q = e|z| < 1$, then*

$$|W_0(z) - M_N(z)| \leq \frac{q^{N+1}}{(N+1)(1-q)}. \quad (9.1)$$

Proof. The integral estimate

$$\log(n!) = \sum_{j=1}^n \log j \geq \int_1^n \log t \, dt = n \log n - n + 1$$

implies $n! \geq (n/e)^n$. Therefore

$$\frac{n^{n-1}}{n!} |z|^n \leq \frac{(e|z|)^n}{n} = \frac{q^n}{n}.$$

For $n \geq N+1$, $1/n \leq 1/(N+1)$, so

$$\sum_{n=N+1}^{\infty} \frac{q^n}{n} \leq \frac{1}{N+1} \sum_{n=N+1}^{\infty} q^n = \frac{q^{N+1}}{(N+1)(1-q)}.$$

□

Alternating signs can make the actual error much smaller on the positive axis, but (9.1) remains valid for complex z and does not rely on cancellation.

9.3 Padé approximants at the origin

A Padé approximant replaces a power series by a rational function whose Taylor expansion matches as many coefficients as possible. The first two diagonal examples are

$$[1/1]_{W_0}(x) = \frac{x}{1+x}, \quad (9.2)$$

$$[2/2]_{W_0}(x) = \frac{x(1 + \frac{4}{3}x)}{1 + \frac{7}{3}x + \frac{5}{6}x^2} = \frac{2x(4x+3)}{5x^2 + 14x + 6}. \quad (9.3)$$

Expanding (9.3) verifies agreement through x^4 . The $[1/1]$ formula is also the global lower bound in (8.1).

Padé approximants are inexpensive and often extend the usefulness of a local series. They do not reproduce the square-root singularity at $-1/e$, however, and their denominator can introduce poles unrelated to W . They are therefore seeds, not universal branch models.

9.4 Puiseux approximants for both branches

Let

$$U_N(p) = \sum_{n=1}^N c_n p^n$$

with the coefficients from (5.6). Then

$$B_{0,N}(x) = -1 + U_N\left(\sqrt{2(1+ex)}\right), \quad (9.4)$$

$$B_{-1,N}(x) = -1 + U_N\left(-\sqrt{2(1+ex)}\right) \quad (9.5)$$

approximate W_0 and W_{-1} , respectively. Since U is analytic at zero,

$$W_k(x) - B_{k,N}(x) = \mathcal{O}\left((1+ex)^{(N+1)/2}\right)$$

after the appropriate branch sign has been selected. The important practical point is that the coefficient table is shared: only the sign of p changes.

9.5 Logarithmic approximants

For $x > 1$, the simplest large-argument approximants are

$$\begin{aligned} A_0^{(0)}(x) &= L - \ell, \\ A_0^{(1)}(x) &= L - \ell + \frac{\ell}{L}, \\ A_0^{(N)}(x) &= L - \ell + \sum_{n=1}^N \frac{P_n(\ell)}{L^n}, \end{aligned}$$

where $L = \log x$ and $\ell = \log L$. For the lower branch, with $T = \log(1/(-x))$ and $\ell = \log T$, use

$$A_{-1}^{(N)}(x) = -T - \ell - \sum_{n=1}^N (-1)^{n-1} \frac{P_n(\ell)}{T^n}. \quad (9.6)$$

These are asymptotic rather than convergent series in general: for a fixed argument, adding terms indefinitely need not improve the result. The optimal truncation depends on the size of L or T . In the ranges plotted in [Figures 9.1](#) and [9.2](#), a few terms already give strong seeds for iteration.

9.6 Inverse-Taylor and Schröder-type corrections

Let

$$h(w) = we^w, \quad z_0 = h(y), \quad \Delta = x - z_0,$$

where y is any initial approximation not equal to -1 . Expanding the local inverse h^{-1} about z_0 gives a systematic family of corrections [\[9\]](#).

Theorem 9.2 (First three inverse-Taylor corrections). *With the notation above, the inverse expansion is*

$$\begin{aligned} W(x) &= y + \frac{e^{-y}}{1+y} \Delta - \frac{e^{-2y}(y+2)}{2(1+y)^3} \Delta^2 \\ &\quad + \frac{e^{-3y}(2y^2+8y+9)}{6(1+y)^5} \Delta^3 + \mathcal{O}(\Delta^4). \end{aligned} \quad (9.7)$$

The first correction is exactly Newton's method for $h(y) - x = 0$. If $y - W(x) = \varepsilon$ and the root is simple, truncation after the term Δ^m leaves an error $\mathcal{O}(\varepsilon^{m+1})$.

Proof. The coefficients are $W^{(j)}(z_0)/j!$, evaluated with the derivative formulas of [Theorem 2.3](#). The linear term is

$$y + \frac{x - ye^y}{e^y(1+y)} = y - \frac{ye^y - x}{e^y(1+y)},$$

which is Newton's formula. Since $\Delta = h(W(x)) - h(y) = \mathcal{O}(\varepsilon)$ near a simple root, the ordinary Taylor remainder $\mathcal{O}(\Delta^{m+1})$ is $\mathcal{O}(\varepsilon^{m+1})$. $\square$

The quadratic truncation is especially attractive: it costs only a few more arithmetic operations than Newton's method and, near a simple root, converts a moderately accurate seed into a third-order approximation. As always, the powers of $(1+y)^{-1}$ warn against using it with a poor seed near the branch point.

9.7 Simple branch-aware seeds

The following elementary choices work well as inputs to one or two corrections. They are not claimed to be minimax approximations.

1. For $W_0(x)$, $x > 0$, use

$$y_0 = \log(1 + x).$$

By (8.1), this lies above the root. It behaves like x near zero and like $\log x$ at infinity.

2. For $W_0(x)$, $-1/e < x < 0$ away from the branch point, use $y_0 = x$. Concavity gives $W_0(x) \leq x$, and the values remain in $(-1, 0)$.
3. For $W_{-1}(x)$, set $T = \log(1/(-x))$ and use

$$y_0 = -T - \log T. \quad (9.8)$$

The proof of [Theorem 8.4](#) shows that this lies above the desired root and below -1 .

4. When $1 + ex$ is small, replace all three generic seeds by the Puiseux approximation. For very large x or very small $-x$, replace them by the corresponding logarithmic expansion.

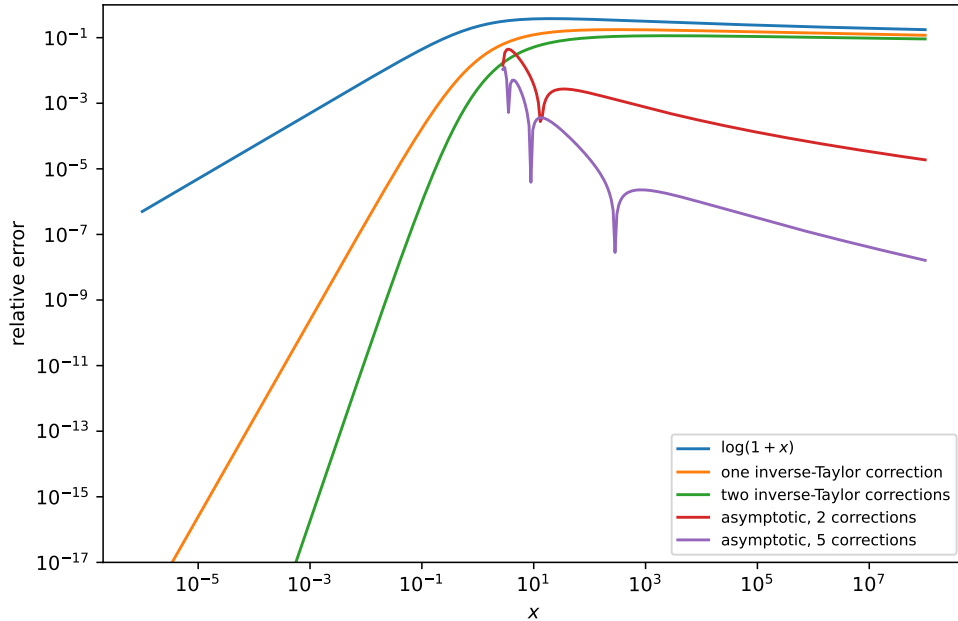


Figure 9.1: Relative errors of several inexpensive approximations to W_0 . Inverse-Taylor correction rapidly improves the global seed $\log(1 + x)$, while logarithmic asymptotics dominate for large arguments.

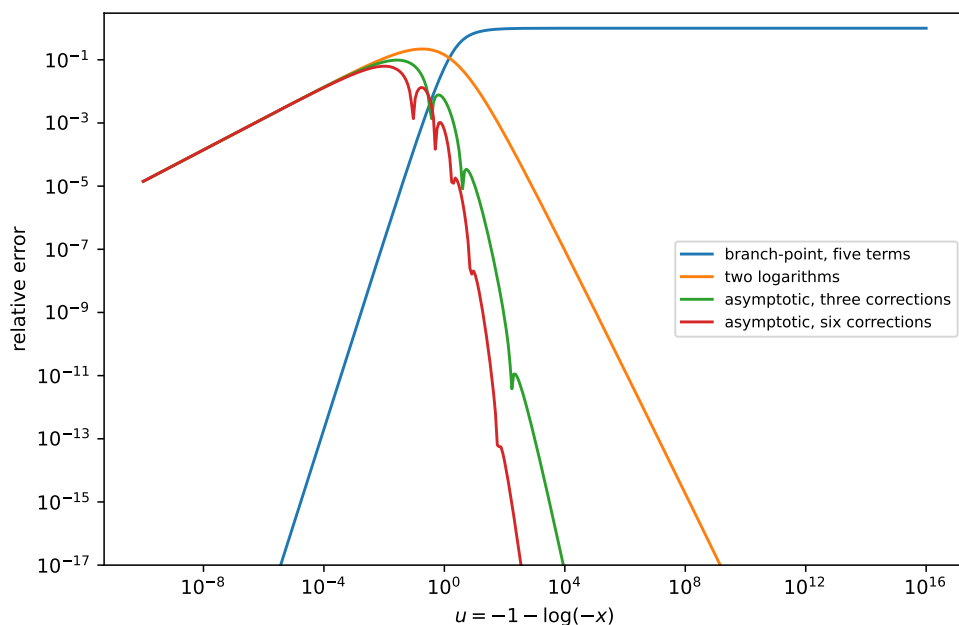


Figure 9.2: Relative errors on W_{-1} , parametrized by $q = -1 - \log(-x)$. The branch-point and logarithmic expansions cover opposite ends of the interval.

9.8 What the approximation literature contributes

The papers in the supplied archive pursue complementary objectives.

Table 9.2: Selected approximation sources in the archive.

source	branches	main emphasis
Barry et al. [2]	W_0, W_{-1}	Explicit piecewise real approximations designed for direct use or as iteration seeds; the paper is notable for treating the lower branch explicitly.
Iacono–Boyd [10]	W_0	Global analytic and spectral approximations, tight large-argument bounds, and an efficient refinement scheme for the principal branch.
Howard [8]	mainly W_0	Geometric and two-point spline constructions whose accuracy can be increased systematically on positive and negative compact intervals.
Howard [9]	W_0, W_{-1}	Generic Schröder-type inverse series built from an arbitrary initial approximating function, with branch-specific convergence conditions.
Lóczi [12]	W_0, W_{-1}	Monotone logarithmic recursion, explicit starting values, and rigorous a-priori iteration counts for guaranteed high precision.

source	branches	main emphasis
Corcino–Corcino– Mező [4]	W_0	C-fractions and associated continued fractions, together with numerical study of their convergents.

The formulas developed in this monograph are deliberately transparent and proof-oriented. For production code requiring a certified number of hundreds or thousands of digits, the sharper enclosures and implementation details in [12] are particularly relevant.

Chapter 10

Branch-aware numerical evaluation

10.1 Why solve the logarithmic equation?

Direct Newton iteration for

$$f(y) = ye^y - x$$

can overflow when y is large and can underflow when x is extremely small on W_{-1} . On a correctly selected real branch, $x/y > 0$, and the equivalent logarithmic equation is

$$g_x(y) = y + \log |y| - \log |x| = 0. \quad (10.1)$$

Its derivatives are

$$g'_x(y) = 1 + \frac{1}{y}, \quad g''_x(y) = -\frac{1}{y^2} < 0.$$

Newton's method becomes

$$\boxed{y_{n+1} = \frac{y_n}{1 + y_n} \left(1 + \log \frac{x}{y_n} \right)}. \quad (10.2)$$

The ratio x/y_n is positive, so the ordinary real logarithm is used on both negative branches as well as on the positive axis.

10.2 Global convergence from elementary starts

The concavity of g_x gives a clean convergence proof.

Theorem 10.1 (Branch-aware logarithmic Newton iteration). *Assume $x \neq -1/e, 0$, and use iteration (10.2).*

(a) *If $x > 0$, set $y_0 = \log(1 + x)$. Then*

$$0 < y_1 \leq W_0(x), \quad y_n \uparrow W_0(x) \quad (n \geq 1).$$

(b) *If $-1/e < x < 0$ and the desired branch is W_0 , set $y_0 = x$. Then*

$$W_0(x) \leq y_{n+1} < y_n < 0, \quad y_n \downarrow W_0(x).$$

(c) If $-1/e < x < 0$ and the desired branch is W_{-1} , let $T = \log(1/(-x))$ and set $y_0 = -T - \log T$. Then

$$y_1 \leq W_{-1}(x) < y_0 < -1, \quad y_n \uparrow W_{-1}(x) \quad (n \geq 1).$$

Proof. We use the elementary tangent geometry of a concave function. If a concave increasing function has a zero r , a Newton step from a point to the right of r lands at or to the left of r ; subsequent Newton steps from the left increase and remain at or to the left of r . If the function is concave and decreasing, a Newton step from the right remains to the right and moves left. These statements follow directly from the fact that the tangent to a concave graph lies above the graph.

For $x > 0$, g_x is increasing and concave on $(0, \infty)$. The bound $W_0(x) \leq \log(1+x) = y_0$ places the start to the right of the root. Also $x > \log(1+x)$, so the explicit formula (10.2) gives $y_1 > 0$. The tangent argument proves the stated bracketing and monotone convergence.

On $(-1, 0)$, $g'_x(y) < 0$, so g_x is decreasing and concave. The bound $W_0(x) \leq x = y_0$ places the start to the right of the root. Each Newton step moves left without crossing the root. The monotone sequence has a limit, and continuity of the Newton map identifies it as the unique zero in $(-1, 0)$.

For the lower branch, g_x is increasing and concave on $(-\infty, -1)$. If $r = W_{-1}(x) = -u$, then $u = T + \log u > T + \log T$, hence $r < -T - \log T = y_0$. Thus the start is to the right of the root. The first Newton step lands at or to its left and therefore remains below -1 . Subsequent steps increase without crossing the root. Their limit must be the unique zero on $(-\infty, -1)$. $\square$

This theorem uses especially simple starts. Lóczi's analysis [12] supplies sharper piecewise starts and explicit uniform bounds on the number of iterations.

10.3 Quadratic convergence and its deterioration at the branch point

Let $r = W_k(x) \neq -1$ and $e_n = y_n - r$. The standard Newton expansion gives

$$e_{n+1} = -\frac{1}{2r(r+1)}e_n^2 + \mathcal{O}(e_n^3). \quad (10.3)$$

Proof. For Newton's method applied to a twice differentiable function with a simple root,

$$e_{n+1} = \frac{g''(r)}{2g'(r)}e_n^2 + \mathcal{O}(e_n^3).$$

Here $g''(r) = -1/r^2$ and $g'(r) = (r+1)/r$, giving (10.3). $\square$

The coefficient becomes unbounded as $r \rightarrow -1$. This is another expression of the square-root conditioning. A branch-point series should therefore be used to obtain a close initial value before Newton or Halley refinement.

10.4 Halley's cubic iteration

Halley's method applied to $f(y) = ye^y - x$ is

$$y_{n+1} = y_n - \frac{y_n e^{y_n} - x}{e^{y_n}(1 + y_n) - \frac{(y_n + 2)(y_n e^{y_n} - x)}{2(1 + y_n)}}. \quad (10.4)$$

At a simple root r ,

$$e_{n+1} = \frac{r^2 + 4r + 6}{12(r+1)^2} e_n^3 + \mathcal{O}(e_n^4). \quad (10.5)$$

Proof. The general Halley formula is

$$y_{n+1} = y_n - \frac{2ff'}{2(f')^2 - ff''}.$$

Substituting $f = ye^y - x$, $f' = e^y(1+y)$, and $f'' = e^y(y+2)$ gives (10.4). Taylor expansion of the general formula at a simple root gives the cubic coefficient

$$\frac{f''(r)^2}{4f'(r)^2} - \frac{f'''(r)}{6f'(r)}.$$

Since $f'''(r) = e^r(r+3)$, simplification yields $(r^2 + 4r + 6)/(12(r+1)^2)$. $\square$

Halley usually reaches machine precision in fewer iterations than Newton after a good seed. It is not a substitute for correct branch selection, and its denominators are again sensitive near $r = -1$.

10.5 A practical real-branch algorithm

The following procedure combines the theory developed above. Thresholds are engineering choices rather than mathematical constants and may be tuned for a particular floating-point format.

Algorithm 10.2 (Piecewise real Lambert W evaluation). Given a real argument x and branch $k \in \{0, -1\}$:

1. Validate the domain. For $k = 0$, require $x \geq -1/e$. For $k = -1$, require $-1/e \leq x < 0$.
2. Return exact endpoint values when $x = 0$ on W_0 or $x = -1/e$ on either branch.
3. Compute $d = 1 + ex$. If d is small, initialize with five to ten terms of the Puiseux series, using $p > 0$ for W_0 and $-p$ for W_{-1} .
4. Otherwise, initialize as follows:
 - $W_0(x)$, $x < 0$: $y = x$;
 - $W_0(x)$, moderate $x > 0$: $y = \log(1+x)$;
 - $W_0(x)$, large x : a few terms of (6.5);
 - $W_{-1}(x)$, moderate x : $y = -T - \log T$;
 - $W_{-1}(x)$, very small $-x$: a few terms of (6.7).
5. Refine with logarithmic Newton or Halley iteration. Stop only when both the proposed correction and a scaled residual satisfy the requested tolerance.
6. Verify that the final value remains in the required range: $y \geq -1$ for W_0 , and $y \leq -1$ for W_{-1} .

10.6 Floating-point details that matter

1. Use `log1p(x)` for $\log(1+x)$ when x is small.
2. Near the branch point, compute $1+ex$ with fused operations when available, or accept the offset $x+1/e$ directly.
3. On W_{-1} , an input may be naturally supplied as $x = -e^{-T}$. Carrying T instead of forming x prevents underflow.
4. Evaluate $\log(x/y)$, not $\log|x| - \log|y|$, when x and y are both negative and their ratio is representable.
5. A small residual near $-1/e$ is not sufficient by itself; combine it with the condition number or with a bracket as in (8.9)–(8.10).

Table 10.1: Iterations of the logarithmic Newton method with the elementary branch-aware starts of [Theorem 10.1](#).

argument	branch	steps for 10^{-15}	final absolute error
10	W_0	4	$1.627e-19$
-0.1	W_0	4	$6.044e-21$
-0.1	W_{-1}	4	$1.28e-24$
$-1/e + 10^{-8}$	W_{-1}	16	$1.483e-21$
-10^{-30}	W_{-1}	2	$1.108e-17$

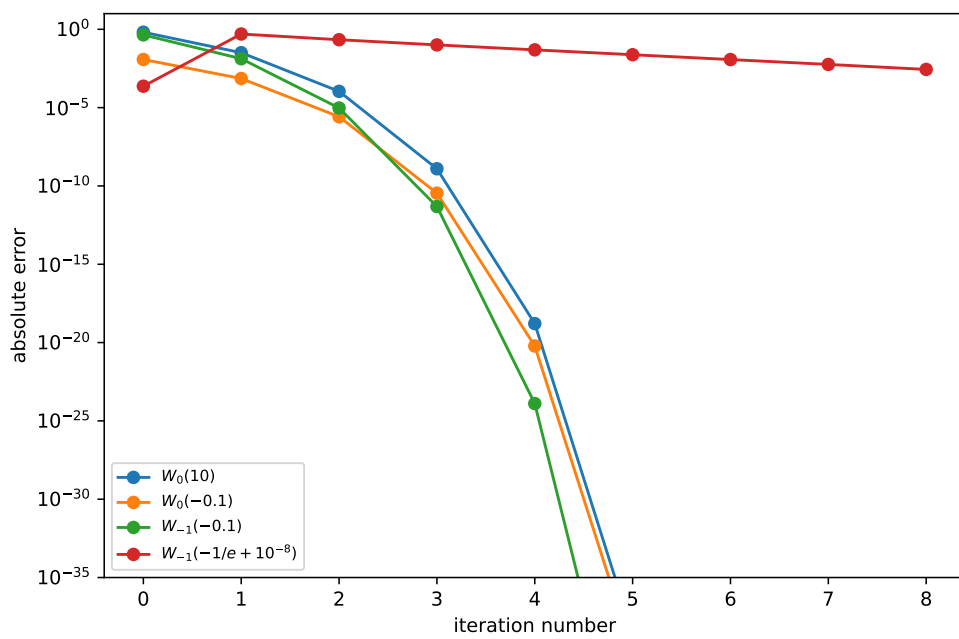


Figure 10.1: Absolute error of logarithmic Newton iteration on representative arguments. Convergence remains quadratic away from the coalescing root at $-1/e$.

Part IV

Applications

Chapter 11

Rooted labeled trees and the tree function

11.1 Cayley's formula

The appearance of n^{n-1} in (4.6) is not accidental. We begin with the classical enumeration theorem.

Theorem 11.1 (Cayley). *There are n^{n-2} trees on a fixed set of $n \geq 2$ labeled vertices. Consequently, there are n^{n-1} rooted labeled trees on those vertices.*

Proof. A Prüfer code associates a sequence of length $n - 2$ over the alphabet $\{1, \dots, n\}$ with each labeled tree. Given a tree, repeatedly remove the smallest-labeled leaf and record its unique neighbor. After $n - 2$ removals, two vertices remain; the recorded labels form the code.

Conversely, given a sequence $(a_1, \dots, a_{n-2})$, at each stage connect the smallest label not appearing in the remaining sequence to the first remaining entry, then delete that entry and the chosen label. Finally connect the last two labels. This reconstructs a tree and reverses the encoding. Thus trees are in bijection with the n^{n-2} sequences. Choosing one of the n vertices as a root multiplies the count by n . $\square$

11.2 Why the functional equation is $T = ze^T$

Let

$$T(z) = \sum_{n \geq 1} t_n \frac{z^n}{n!}$$

be the exponential generating function for rooted labeled trees. Remove the root. What remains is an unordered set of rooted labeled trees, one for each neighbor of the old root. In labeled enumeration, an unordered set of objects with exponential generating function T has generating function e^T ; the extra root contributes a factor z . Hence

$$T(z) = ze^{T(z)}. \tag{11.1}$$

For completeness, the coefficient mechanism behind the exponential is this: choosing j unordered components contributes $T(z)^j/j!$, and summing over $j \geq 0$ gives $\sum T^j/j! = e^T$. The

labels are distributed among the components by the multinomial coefficients already built into exponential generating functions.

Equation (11.1) is equivalent to

$$-T(z)e^{-T(z)} = -z,$$

after multiplying by -1 . Therefore

$$T(z) = -W_0(-z). \quad (11.2)$$

Lagrange inversion now gives

$$t_n = n^{n-1},$$

recovering the rooted form of Cayley's formula analytically.

11.3 Forests and powers of W

An unordered forest of m rooted labeled trees has exponential generating function $T(z)^m/m!$. Applying Lagrange inversion to $T = ze^T$ gives

$$[z^n]T(z)^m = \frac{m}{n} [t^{n-m}]e^{nt} = \frac{m}{n} \frac{n^{n-m}}{(n-m)!}.$$

Thus the number of forests on n labeled vertices with exactly m rooted components is

$$n! [z^n] \frac{T(z)^m}{m!} = \binom{n}{m} m n^{n-m-1}. \quad (11.3)$$

This formula can also be read as: choose the m roots, then form a forest oriented toward them. It illustrates why the power series (4.3) is combinatorially meaningful rather than merely a formal curiosity.

11.4 The singularity and coefficient growth

The dominant positive singularity of T is $z = 1/e$. Substituting $x = -z$ in the branch-point expansion gives, with $p = \sqrt{2(1-ez)}$,

$$T(z) = 1 - p + \frac{1}{3}p^2 - \frac{11}{72}p^3 + \cdots. \quad (11.4)$$

The square-root singularity is responsible for the power $n^{-3/2}$ in the asymptotic size of the exponential-generating-function coefficients. Directly from Stirling's formula,

$$\frac{n^{n-1}}{n!} \sim \frac{e^n}{\sqrt{2\pi} n^{3/2}}. \quad (11.5)$$

Thus the geometry of the inverse at $-1/e$, the convergence radius $1/e$, and the asymptotic enumeration of trees are three views of the same phenomenon.

Chapter 12

Delay equations and exponential dynamics

12.1 A scalar delay differential equation

Consider

$$y'(t) = a y(t - \tau), \quad \tau > 0. \quad (12.1)$$

Looking for exponential modes $y(t) = e^{\lambda t}$ gives

$$\lambda e^{\lambda t} = a e^{\lambda(t-\tau)}, \quad \lambda e^{\lambda \tau} = a.$$

Multiplying by τ ,

$$\lambda_k = \frac{1}{\tau} W_k(a\tau), \quad k \in \mathbb{Z}. \quad (12.2)$$

A delay equation has infinitely many characteristic exponents because W has infinitely many complex branches. The complex branch structure is not an ornament here; it is the spectrum.

12.2 The affine linear delay equation

The more general equation

$$y'(t) = \alpha y(t) + \beta y(t - \tau) \quad (12.3)$$

has characteristic equation

$$\lambda = \alpha + \beta e^{-\lambda \tau}.$$

Let $u = (\lambda - \alpha)\tau$. Then

$$u e^u = \beta \tau e^{-\alpha \tau},$$

so

$$\lambda_k = \alpha + \frac{1}{\tau} W_k(\beta \tau e^{-\alpha \tau}). \quad (12.4)$$

Every mode follows by a one-line reduction to the inverse relation. Determining stability still requires examining the real parts of all relevant branches.

12.3 How the two real branches enter a delay spectrum

For the negative-feedback equation

$$y'(t) = -ay(t - \tau), \quad a > 0,$$

we have $\lambda_k = \tau^{-1}W_k(-a\tau)$. If $0 < a\tau < 1/e$, two characteristic exponents are real:

$$\lambda_0 = \frac{1}{\tau}W_0(-a\tau), \quad \lambda_{-1} = \frac{1}{\tau}W_{-1}(-a\tau).$$

The W_{-1} mode is more negative and therefore decays faster. When $a\tau > 1/e$, these two real modes merge and continue as a complex-conjugate pair on adjacent sheets.

The loss of stability occurs later. Put $\lambda = i\omega$ in $\lambda = -ae^{-\lambda\tau}$. Taking absolute values gives $\omega = a$. Matching arguments at the first crossing gives

$$\frac{\pi}{2} = \pi - a\tau, \quad a\tau = \frac{\pi}{2}.$$

Thus the branch point $a\tau = 1/e$ marks the transition from real to oscillatory dominant modes, while $a\tau = \pi/2$ marks the first imaginary-axis crossing. These are distinct dynamical events.

12.4 Falling with linear drag: an inversion in time

Let an object start from rest and approach terminal speed v_∞ with linear-drag time scale $1/k$. Its distance after time t is

$$s(t) = v_\infty t - \frac{v_\infty}{k}(1 - e^{-kt}).$$

Set $u = kt$ and $D = ks/v_\infty$. Then

$$D = u - 1 + e^{-u}.$$

Writing $c = D + 1$ and $v = c - u$, we obtain

$$ve^{-v} = e^{-c}, \quad -v = W_k(-e^{-c}),$$

so

$$t = \frac{D + 1 + W_k(-e^{-D-1})}{k}. \quad (12.5)$$

For $D > 0$, the principal branch gives the positive physical time. The lower branch produces a nonpositive algebraic solution and is rejected by the initial-value problem. At $D = 0$, the two branches coalesce and give $t = 0$.

Chapter 13

Kinetics, epidemics, and ecological models

13.1 A universal logarithm-plus-linear equation

Many integrated rate laws reduce to

$$x + a \log x = b, \quad x > 0, \quad (13.1)$$

where $a \neq 0$. Dividing by a and exponentiating gives

$$xe^{x/a} = e^{b/a},$$

so

$$x = aW_k\left(\frac{1}{a}e^{b/a}\right). \quad (13.2)$$

The branch and even the sign of the argument depend on a . For $a > 0$ the argument is positive and only W_0 is real; for $a < 0$, two real solutions may occur.

13.2 Integrated Michaelis–Menten kinetics

Let $S(t) > 0$ be substrate concentration and suppose

$$\frac{dS}{dt} = -\frac{V_{\max}S}{K_M + S}, \quad S(0) = S_0 > 0. \quad (13.3)$$

Separating variables gives

$$\left(1 + \frac{K_M}{S}\right) dS = -V_{\max} dt,$$

and hence

$$S + K_M \log S = S_0 + K_M \log S_0 - V_{\max} t.$$

Put $s = S/K_M$, $s_0 = S_0/K_M$. Constants involving $\log K_M$ cancel, leaving

$$s + \log s = s_0 + \log s_0 - \frac{V_{\max}}{K_M} t.$$

Exponentiation yields

$$se^s = s_0 \exp\left(s_0 - \frac{V_{\max}}{K_M} t\right),$$

so the exact solution is

$$S(t) = K_M W_0\left(\frac{S_0}{K_M} \exp\left(\frac{S_0 - V_{\max} t}{K_M}\right)\right). \quad (13.4)$$

The argument is positive, so the principal branch is forced by positivity. This example, and more complicated modified rate laws, are developed in the biochemical review by Goličnik [7].

13.3 The final-size equation of an SIR epidemic

In normalized SIR variables, let $s + i + r = 1$ and

$$\dot{s} = -\mathcal{R}_0 s i, \quad \dot{r} = i.$$

Eliminating time gives

$$\frac{ds}{dr} = -\mathcal{R}_0 s, \quad \log \frac{s}{s_0} = -\mathcal{R}_0 (r - r_0).$$

At the end of the epidemic, $i_\infty = 0$ and $r_\infty = 1 - s_\infty$. Therefore

$$s_\infty = s_0 \exp\{-\mathcal{R}_0(1 - s_\infty - r_0)\}.$$

Multiplication by $-\mathcal{R}_0 e^{-\mathcal{R}_0 s_\infty}$ gives

$$s_\infty = -\frac{1}{\mathcal{R}_0} W_k\left(-\mathcal{R}_0 s_0 e^{-\mathcal{R}_0(1-r_0)}\right). \quad (13.5)$$

The limiting idealization $s_0 = 1$, $r_0 = 0$ makes branch selection particularly instructive:

$$s_\infty = -\frac{1}{\mathcal{R}_0} W_k(-\mathcal{R}_0 e^{-\mathcal{R}_0}).$$

If $\mathcal{R}_0 > 1$, then $W_{-1}(-\mathcal{R}_0 e^{-\mathcal{R}_0}) = -\mathcal{R}_0$, which gives the trivial solution $s_\infty = 1$. The physically relevant macroscopic outbreak is furnished by W_0 , giving $s_\infty < 1/\mathcal{R}_0$. If $0 < \mathcal{R}_0 < 1$, the value $-\mathcal{R}_0$ belongs instead to the principal range, so W_0 gives the no-outbreak solution and the lower branch would give $s_\infty > 1$, which is inadmissible. The branch switch mirrors the epidemic threshold.

Lambert W appears in many related ecological and evolutionary problems, including marginal-value optimization, population growth rates, mate search, and fertilization kinetics; Lehtonen gives a broad model-centered survey [11].

Chapter 14

Diode circuits and physical transcendental equations

14.1 The generalized diode equation

For a diode with saturation current I_s , series resistance R , applied voltage V , and thermal-voltage factor $a = nV_T > 0$, the Shockley law with series resistance is

$$I = I_s \left(\exp \frac{V - RI}{a} - 1 \right). \quad (14.1)$$

Set $J = I + I_s$. Since $I = J - I_s$,

$$J = I_s \exp \left(\frac{V + RI_s - RJ}{a} \right).$$

Thus

$$\frac{RJ}{a} \exp \left(\frac{RJ}{a} \right) = \frac{RI_s}{a} \exp \left(\frac{V + RI_s}{a} \right),$$

and

$$I = \frac{a}{R} W_0 \left(\frac{RI_s}{a} \exp \frac{V + RI_s}{a} \right) - I_s. \quad (14.2)$$

The argument is positive, so the current is single-valued on W_0 . This closed form creates a smooth analytical bridge between exponential diode behavior and the resistor-dominated regime. Banwell uses it as the basis for broader bipolar-transistor circuit analysis [1].

14.2 Wien's displacement equation and a discarded branch

The maximum of the wavelength form of Planck's blackbody spectrum leads to

$$5(1 - e^{-x}) = x. \quad (14.3)$$

Let $y = x - 5$. Then

$$ye^y = -5e^{-5},$$

so

$$x = 5 + W_k(-5e^{-5}). \quad (14.4)$$

Because the argument lies in $(-1/e, 0)$, there are two real branch values. The lower branch is exact:

$$W_{-1}(-5e^{-5}) = -5,$$

which gives $x = 0$, a boundary root introduced by the algebraic stationarity equation. The nonzero physical maximum is

$$x = 5 + W_0(-5e^{-5}) = 4.965114231744 \dots$$

This is a model example of why both branches must be written down before physical constraints are imposed.

14.3 A generic saturation equation

Equations of the form

$$A(1 - e^{-bx}) = cx, \quad A, b, c > 0, \quad (14.5)$$

occur in transport, charging, response saturation, and finite-time relaxation. Put $d = Ab/c$ and $u = bx$. Then

$$d(1 - e^{-u}) = u.$$

The same algebra as in the Wien example gives

$$u = d + W_k(-de^{-d}), \quad x = \frac{d + W_k(-de^{-d})}{b}. \quad (14.6)$$

One branch always gives the trivial root $u = 0$; depending on whether $d > 1$, the other branch gives a positive nontrivial response. The threshold $d = 1$ is exactly the coalescence point of the two real branches.

Chapter 15

Prime-number inversion

15.1 Inverting $y \log y = x$

The elementary inverse

$$y \log y = x \iff y = \exp(W_0(x)) = \frac{x}{W_0(x)}$$

turns classical estimates involving $n \log n$ into estimates involving W . This is the basic observation behind Visser's treatment of primes [13].

Theorem 15.1 (A Lambert W upper bound for the prime-counting function). *Assume the classical Rosser inequality $p_n > n \log n$ for $n \geq 1$. Then for $x \geq 0$,*

$$\pi(x) < \exp(W_0(x)) = \frac{x}{W_0(x)}, \quad (15.1)$$

with the quotient interpreted by continuity at $x = 0$.

Proof. For $x < 2$, the assertion is checked directly. For $x \geq 2$, let $n = \pi(x)$, so $p_n \leq x$. The assumed prime bound gives

$$x \geq p_n > n \log n.$$

The function $y \mapsto y \log y$ is increasing for $y \geq 1$. Its inverse at x is $\exp(W_0(x))$, so $n < \exp(W_0(x))$. Finally, $\exp(W_0(x)) = x/W_0(x)$ follows from the defining relation. $\square$

15.2 Why the lower branch estimates the n th prime

The rough relation $n \approx x/\log x$ is inverted as follows. Put $u = \log x$, so $x = e^u$. Then

$$n = \frac{e^u}{u} \iff (-u)e^{-u} = -\frac{1}{n}.$$

The large solution has $u > 1$, so $-u < -1$ and the correct branch is W_{-1} :

$$x = -nW_{-1}(-1/n). \quad (15.2)$$

The principal branch would give a small u and therefore the wrong inverse.

Using known explicit inequalities for $\pi(x)$, Visser proves, for example,

$$p_n < -nW_{-1}(-1/n), \quad n \geq 4. \quad (15.3)$$

The Lambert form cleanly separates the elementary inversion from the number-theoretic input.

A refined approximation to the prime number theorem is

$$n \approx \frac{x}{\log x - 1}.$$

If $u = \log x - 1$, then

$$n = \frac{e^{u+1}}{u}, \quad (-u)e^{-u} = -\frac{e}{n},$$

and hence

$$x \approx -nW_{-1}(-e/n). \quad (15.4)$$

The first terms of [Theorem 6.4](#) give

$$-W_{-1}(-e/n) = \log n + \log \log n - 1 + o(1),$$

so the three familiar leading terms of the Cesàro–Cipolla expansion appear automatically. Higher terms of the Lambert expansion provide a systematic closed-form refinement, though they do not by themselves replace the deeper number-theoretic corrections needed for the full asymptotic series of p_n .

Appendix A

Formula compendium and coefficient tables

A.1 Real branch summary

Table A.1: The two real branches at a glance.

property	W_0	W_{-1}
domain	$[-1/e, \infty)$	$[-1/e, 0)$
range	$[-1, \infty)$	$(-\infty, -1]$
monotonicity	increasing	decreasing
curvature	concave	convex for $-2 < W < -1$, then concave for $W < -2$; see Chapter 2
value at branch point	-1	-1
origin behavior	$W_0(x) = x - x^2 + \dots$	$W_{-1}(x) \rightarrow -\infty$ logarithmically
far-end behavior	$W_0(x) \sim \log x - \log \log x$	$W_{-1}(x) \sim \log(-x) - \log[-\log(-x)]$
branch-point	$-1 + U(p)$	$-1 + U(-p)$
sign		

In the curvature row, the argument interval for convexity is $(-1/e, -2/e^2)$, and the interval for concavity is $(-2/e^2, 0)$. The compact wording in the table is intended only as a memory aid.

A.2 Derivative polynomials

The recurrence from [Theorem 2.3](#) produces

$$W^{(n)}(x) = \frac{e^{-nW(x)} P_n(W(x))}{(1 + W(x))^{2n-1}}.$$

The first eight polynomials are

$$\begin{aligned}
P_1(w) &= 1, \\
P_2(w) &= -w - 2, \\
P_3(w) &= 2w^2 + 8w + 9, \\
P_4(w) &= -6w^3 - 36w^2 - 79w - 64, \\
P_5(w) &= 24w^4 + 192w^3 + 622w^2 + 974w + 625, \\
P_6(w) &= -120w^5 - 1200w^4 - 5126w^3 - 11758w^2 - 14543w - 7776, \\
P_7(w) &= 720w^6 + 8640w^5 + 45756w^4 + 137512w^3 \\
&\quad + 248250w^2 + 255828w + 117649, \\
P_8(w) &= -5040w^7 - 70560w^6 - 445572w^5 - 1651480w^4 \\
&\quad - 3892430w^3 - 5846760w^2 - 5187775w - 2097152.
\end{aligned}$$

The constants $P_n(0) = (-n)^{n-1}$ reproduce the Maclaurin derivatives.

A.3 Branch-point coefficients

For quick reference, the first coefficients in $U(p) = \sum_{n \geq 1} c_n p^n$ are

n	c_n	n	c_n
1	1	6	$-221/8505$
2	$-1/3$	7	$680863/43545600$
3	$11/72$	8	$-1963/204120$
4	$-43/540$	9	$226287557/37623398400$
5	$769/17280$	10	$-5776369/1515591000$

Use $W_0 = -1 + U(p)$ and $W_{-1} = -1 + U(-p)$.

A.4 A compact identity list

For a consistent branch and away from singular points,

$$\begin{aligned}
 W e^W &= x, \\
 e^W &= \frac{x}{W}, \\
 W &= \log x - \log W, \\
 W' &= \frac{W}{x(1+W)} = \frac{e^{-W}}{1+W}, \\
 xW' &= \frac{W}{1+W}, \\
 \frac{d}{dx} \left(W + \frac{1}{2}W^2 \right) &= \frac{W}{x}, \\
 \frac{d}{dx} \left(\frac{x}{W} \right) &= \frac{1}{1+W}, \\
 \frac{d}{dx} \log |W| &= \frac{1}{x(1+W)}.
 \end{aligned}$$

The logarithmic identity must be interpreted with compatible complex branches; on the real branches it is safest to write $W + \log |W| = \log |x|$.

Appendix B

How the attached archive was used

The supplied archive combines foundational papers, recent approximation work, application surveys, reference exports, and a computer-algebra notebook. The following guide records the role of each item and helps distinguish primary sources from convenient secondary summaries.

Table B.1: Guide to the files in the supplied `Lambert-W.zip`.

file or group	use in this article
<code>corless1996.pdf</code> ; <code>On-the-Lambert-W-Function.pdf</code>	Two renderings of the canonical Corless–Gonnet–Hare–Jeffrey–Knuth paper. They supplied the standard branch notation, complex continuation, derivative structure, Maclaurin and branch-point series, all-branch asymptotics, symbolic integration viewpoint, and historical context.
<code>DLMF_4.13 Lambert W-Function.pdf</code>	Authoritative reference summary for branches, special values, expansions, integrals, differential identities, and the Wright omega and tree functions.
<code>barry2000.pdf</code>	Early explicit approximations covering the separate real portions of W_0 and, importantly, W_{-1} .
<code>chatzigeorgiou2013.pdf</code>	The global square-root-plus-linear bounds for $W_{-1}(-e^{-1-q})$, reproved in Theorem 8.6 .
<code>corcino2018.pdf</code>	The Fourier–logarithm integral and the regular and associated continued fractions for the principal branch.
<code>iacono2017.pdf</code>	Global principal-branch approximations, spectral approximation issues, large-argument bounds, and efficient iterative refinement.
<code>10.1016/j.amc.2022.127406.pdf</code>	Lóczy’s guaranteed high-precision logarithmic recursion, monotone starts, and explicit convergence estimates for both real branches.
<code>_Manuscript.pdf_.pdf</code>	Howard’s geometric and spline-based construction of systematically improvable approximations, especially for the principal branch.
<code>appliedmath-05-00066.pdf</code>	Howard’s 2025 Schröder-type inverse-series framework for both real branches. It motivated the branch-neutral inverse-Taylor treatment in Section 9.6 .
<code>banwell2000.pdf</code>	Exact generalized-diode and bipolar-transistor formulas using W , represented here by (14.2) .

file or group	use in this article
10.1016@j.bej.2012.01.010.pdf	Goličnik's review of biochemical kinetics, including integrated enzyme rate laws and related transport-limited systems.
lehtonen2016.pdf	Broad ecological and evolutionary survey, including epidemic, population, optimization, and search models.
visser2018.pdf	Prime-counting and n th-prime bounds and asymptotics expressed through W_0 and W_{-1} .
ProductLog.nb; ProductLog.pdf	A Wolfram notebook and rendered output illustrating symbolic identities, series, and the computer-algebra notation <code>ProductLog</code> . They were used as computational cross-checks, not as the source of proofs.
Lambert W-Function - from Wolfram MathWorld.pdf	Secondary reference for historical notes, identities, and special values.
Lambert W function - Wikipedia.pdf; Lambert W-function.txt	Snapshot and source text of a general encyclopedia article. These were used only as navigation aids and cross-checks; primary papers and DLMF take precedence when precision matters.

Appendix C

Reproducing the numerical experiments

The ZIP accompanying this article contains `lambert_w_experiments.py`. The program is deterministic and uses no random data. It requires Python 3 together with `mpmath`, `numpy`, and `matplotlib`. From the article directory, run

```
python3 lambert_w_experiments.py
```

to regenerate the PDF figures and the two small LaTeX tables in the `generated` directory.

C.1 What the program checks

The script performs the following independent numerical tasks.

1. It evaluates W_0 and W_{-1} with 50 decimal digits using `mpmath` and plots the two real branches.
2. It compares one- and five-term branch-point expansions with high-precision reference values over many scales of $1 + ex$.
3. It verifies the condition-number law $1/|1 + W| \sim 1/\sqrt{2(1 + ex)}$.
4. It evaluates the asymptotic polynomials $P_1, \dots, P_6$ on both real branches and plots relative errors.
5. It applies one and two inverse-Taylor corrections to elementary seeds.
6. It tests the monotone logarithmic Newton iteration on positive, negative principal, lower-branch, and near-branch-point examples.
7. It writes the numerical value and iteration-count tables included in the book.

The plotted curves are evidence and diagnostics, not substitutes for the proofs. In particular, high-precision reference values were used to detect sign mistakes in the odd powers of the W_{-1} branch-point series and in the alternating lower-branch asymptotic corrections.

C.2 Compiling the document

A typical TeX installation can compile the source with

```
latexmk -pdf -interaction=nonstopmode lambert_w_monograph.tex
```

The distributed archive already contains the rendered PDF, all generated figures, and the generated table fragments, so rerunning the Python program is optional.

Appendix D

Further directions

The ordinary Lambert function is the first member of a larger family of inverse functions. Equations such as

$$xe^x + rx = a,$$

products of shifted linear factors times an exponential, and matrix or operator analogues lead to generalized Lambert functions. Many techniques in this article survive: locate critical points of the forward map, choose local coordinates adapted to those critical points, use Lagrange inversion for local series, take logarithms for remote asymptotics, and preserve every real inverse branch during algebraic reduction.

Several questions remain attractive even for the classical function. Uniform approximations that are simultaneously elementary, branch-aware, near-minimax, and rigorously certified continue to be refined. Complex interval algorithms must manage branch cuts and coalescing roots without losing correct enclosures. In applications, an explicit W formula often exposes previously hidden multiplicity or threshold behavior; deciding which branch is physically realized may require stability, initial data, or variational arguments beyond the algebra itself.

Bibliography

- [1] T. C. Banwell, “Bipolar transistor circuit analysis using the Lambert W -function,” *IEEE Transactions on Circuits and Systems I*, vol. 47, no. 11, pp. 1621–1633, 2000.
- [2] D. A. Barry, J.-Y. Parlange, L. Li, H. Prommer, C. J. Cunningham, and F. Stagnitti, “Analytical approximations for real values of the Lambert W -function,” *Mathematics and Computers in Simulation*, vol. 53, pp. 95–103, 2000.
- [3] I. Chatzigeorgiou, “Bounds on the Lambert function and their application to the outage analysis of user cooperation,” *IEEE Communications Letters*, vol. 17, no. 8, pp. 1505–1508, 2013.
- [4] C. B. Corcino, R. B. Corcino, and I. Mező, “Continued fraction expansions for the Lambert W function,” *Aequationes Mathematicae*, 2018, doi:10.1007/s00010-018-0559-2.
- [5] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, “On the Lambert W function,” *Advances in Computational Mathematics*, vol. 5, pp. 329–359, 1996.
- [6] NIST Digital Library of Mathematical Functions, “Section 4.13: Lambert W -function,” F. W. J. Olver et al., eds., National Institute of Standards and Technology.
- [7] M. Goličnik, “On the Lambert W function and its utility in biochemical kinetics,” *Biochemical Engineering Journal*, vol. 63, pp. 116–123, 2012.
- [8] R. M. Howard, “Analytical approximations for the principal branch of the Lambert W function,” manuscript, 2022; copy supplied in the source archive.
- [9] R. M. Howard, “On Schröder-type series expansions for the Lambert W function,” *Applied-Math*, vol. 5, article 66, 2025, doi:10.3390/appliedmath5020066.
- [10] R. Iacono and J. P. Boyd, “New approximations to the principal real-valued branch of the Lambert W -function,” *Advances in Computational Mathematics*, vol. 43, pp. 1403–1436, 2017.
- [11] J. Lehtonen, “The Lambert W function in ecological and evolutionary models,” *Methods in Ecology and Evolution*, 2016, doi:10.1111/2041-210X.12568.
- [12] L. Lóczi, “Guaranteed- and high-precision evaluation of the Lambert W function,” *Applied Mathematics and Computation*, vol. 433, article 127406, 2022.
- [13] M. Visser, “Primes and the Lambert W function,” *Mathematics*, vol. 6, article 56, 2018.